

Calibrating the radar

Hopefully.

Overview & Purpose

- To develop an 'intelligent' Gain Scheduler Algorithm.
 - True radar characteristics are required. Simply, an unfiltered, raw output of Radar is needed.
 - Prior work has already made significant progress; all theoretical work is done, only final implementation is needed
- What is needed:
 - Radar + Capture Setup
 - Corner reflector (or object with known reflectivity and RCS)
 - A long, straight, clear of obstructions field or plane

Steps

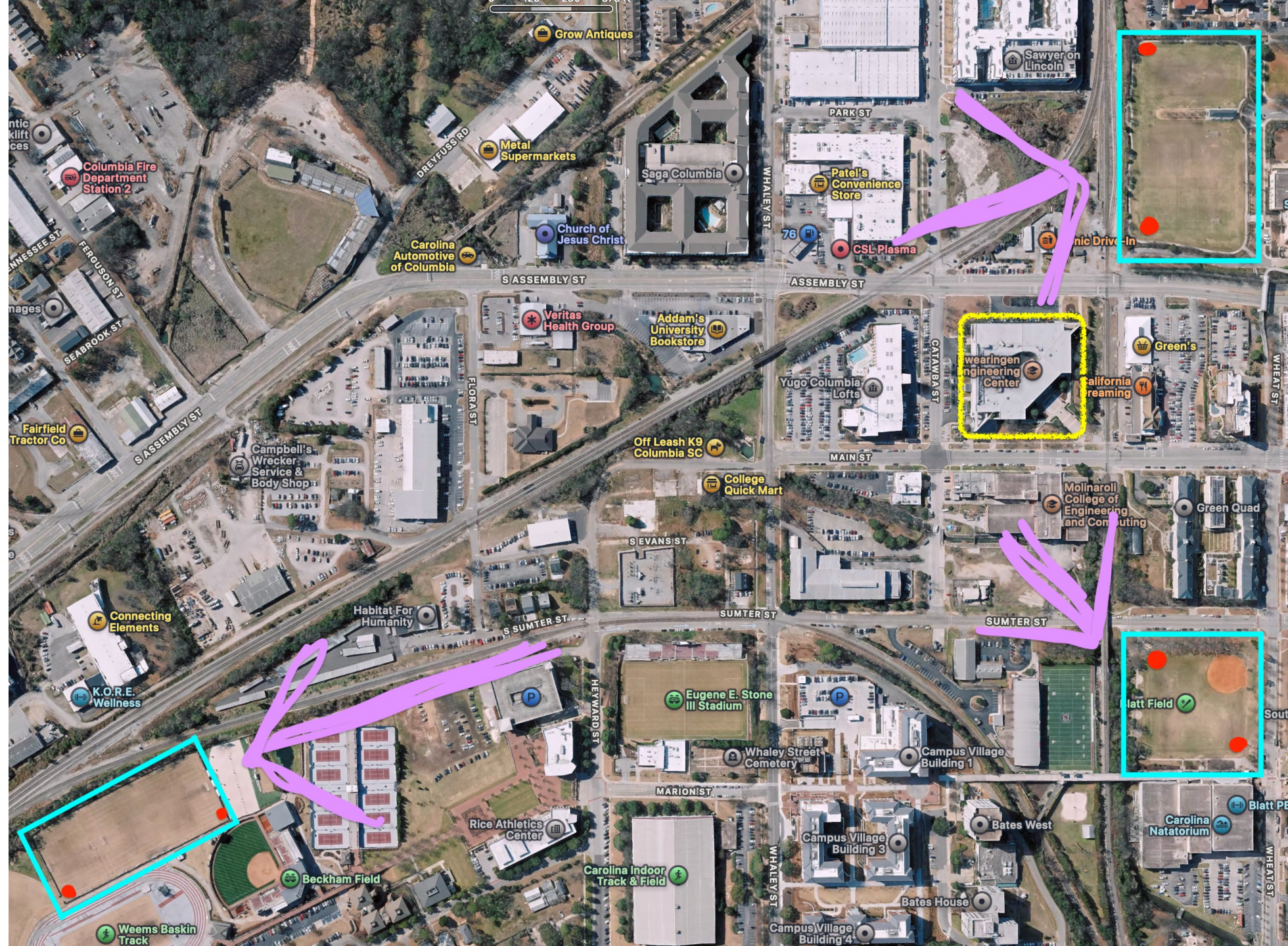
- Setup radar in lab with Pi (testing to ensure all functions are working and to find bugs beforehand)
- Set all STC values to 0 or their lowest values. (Sensitivity Time Control)
- Copy command list attached at end of ppt.
 - Just changes pulse length & gain. Keeps single range.
- Turn off pi and radar and prep battery

Steps

- Measure the corner reflector. All sides should be length 'a'.
 - Put a in this equation to calculate the minimum distance the reflector should be from radar. $L = 2 \cdot a^2 / 0.032^3$ where a is measured in meters
- Place the reflector on the ground(?) at the distance set away from the radar
- Once the gain sweep is complete, repeat but place the reflector at the furthest point from the radar(bound by land available)
 - be certain there are not other reflectors near the reflector, else 'merging' of objects may occur
- Let the gain ramp cycle occur again, then move the object to the middle point between the furthest and closest points.

3 possible locations

Note the red dots are possible locations to place the radar and reflector



code

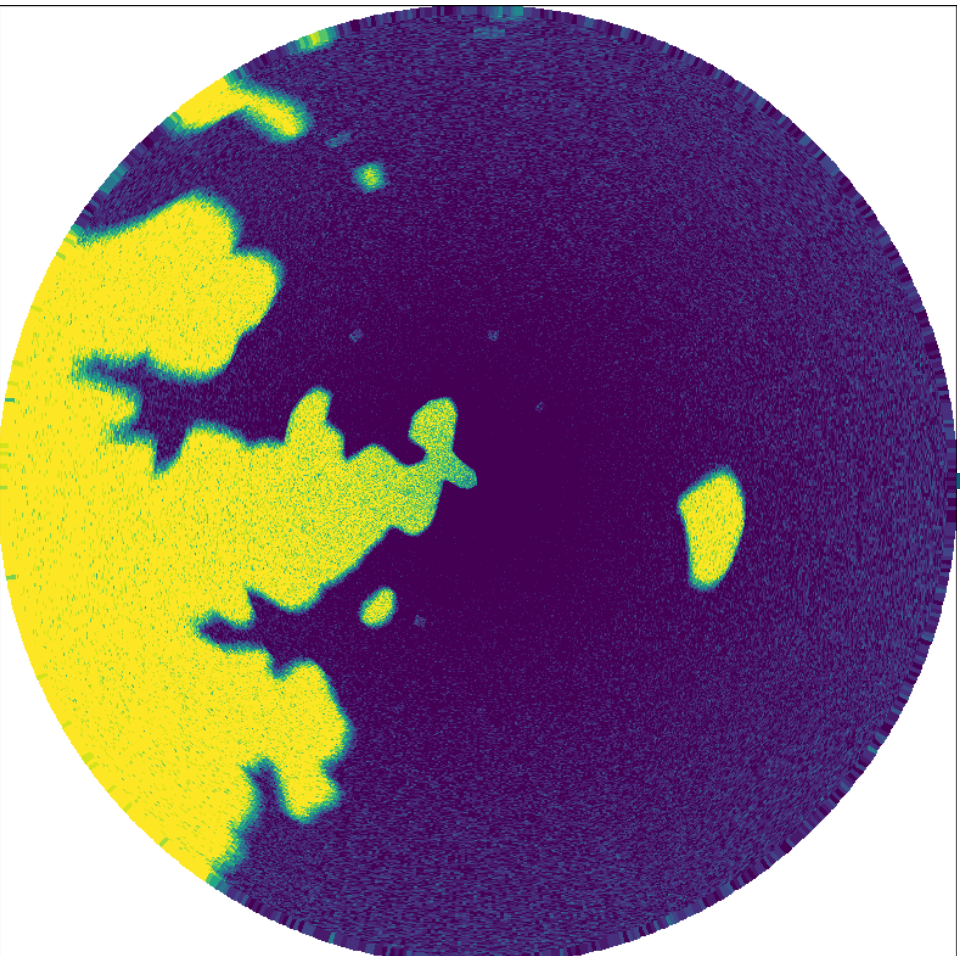
- Copy command list attached at end of ppt.
 - Just changes pulse length & gain. Keeps single range.
- At slow speed, Time is $100 * 10 * 3 * 3 * 2 = 18,000\text{s} = 5\text{hr}$
- At high speed, Time is $100 * 10 * 3 * 3 = 9,000\text{s} = 2.5\text{hr}$
- With 5 samples instead of 10, $100 * 5 * 3 * 3 = 4,500\text{s} = 1.66\text{hr}$

Things that we need to consider

- Objects within the beam width of the radar.
 - Radar beam width is 3.9 deg. We need to calculate that distance out and just ensure nothing is in that 'band'
- Ensuring STC is off. If STC is enabled, the radar will apply a 'filtering' curve to the data.

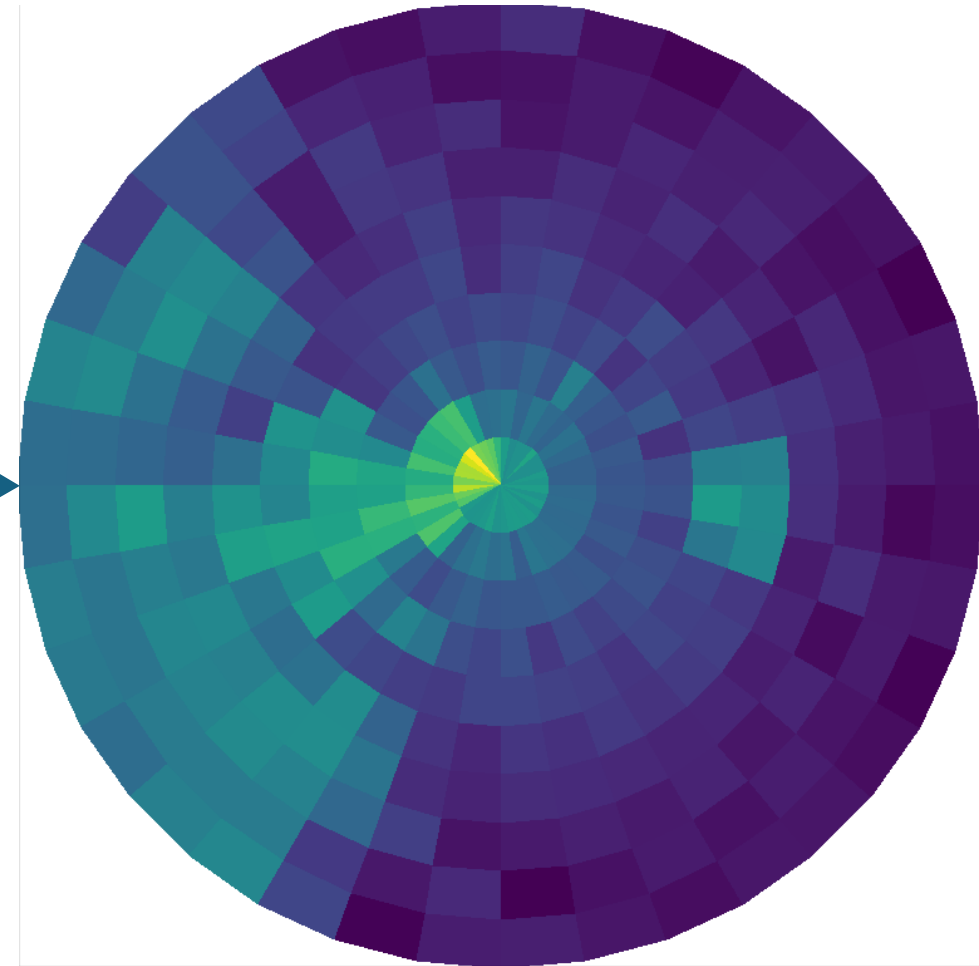
Radar Simulations - Explainer

Full quality radar output – 868 bins, 720 rays



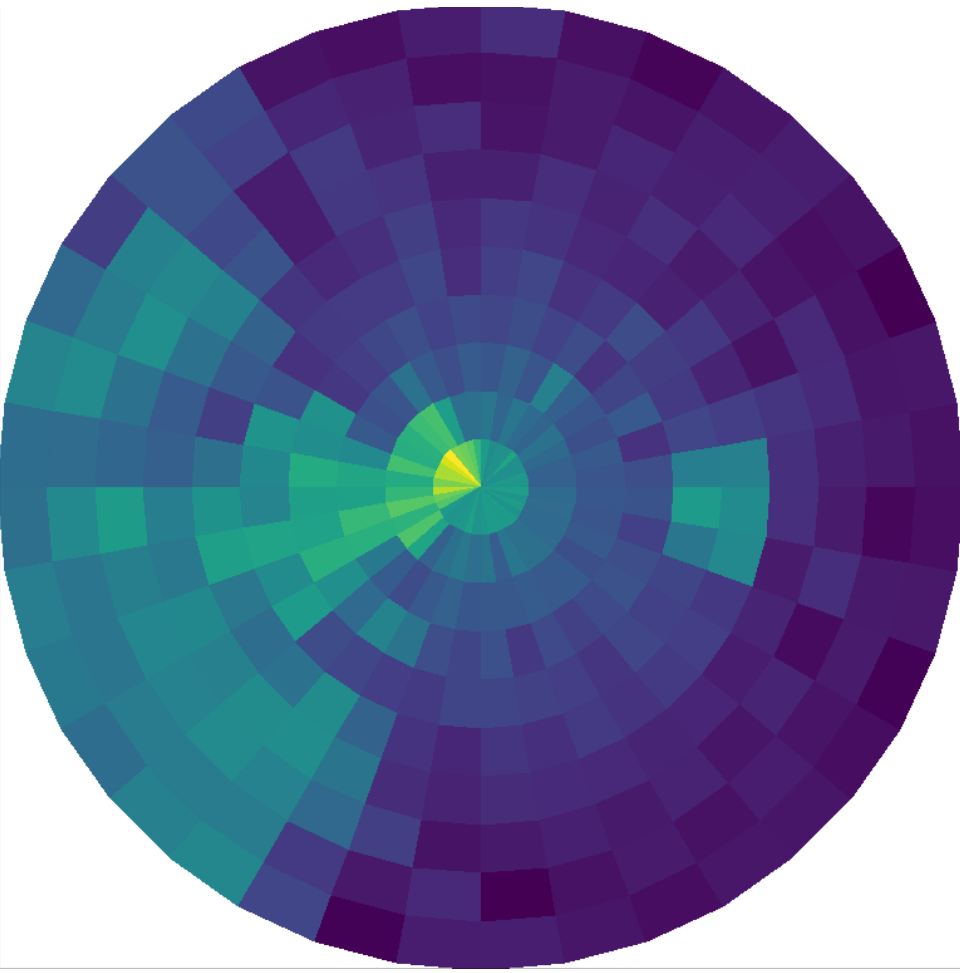
SIMPLIFY

Simple Simulation – 10 bins, 36 rays, 4 bit



Radar Simulations - Explainer

Simplified model – 10 bins, 36 rays, 4 bit



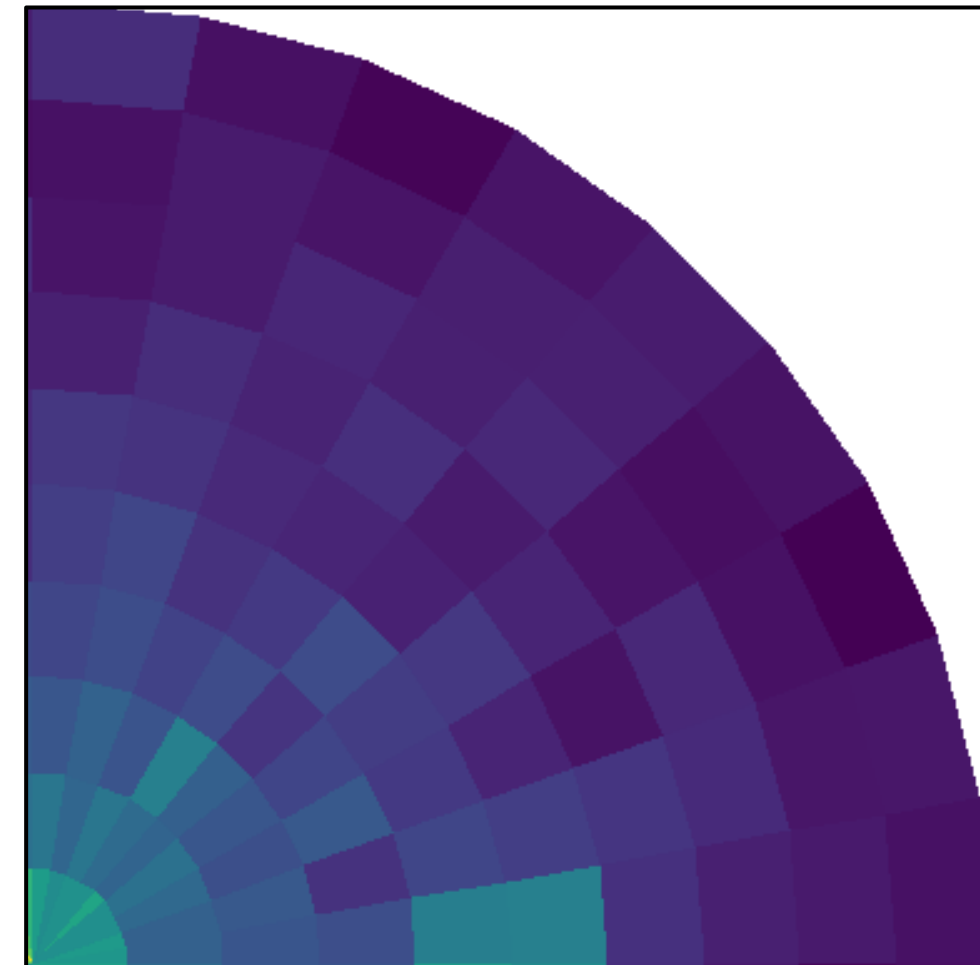
SIMPLIFY

Simple Simulation – 10 bins, 9 rays, 4 bit



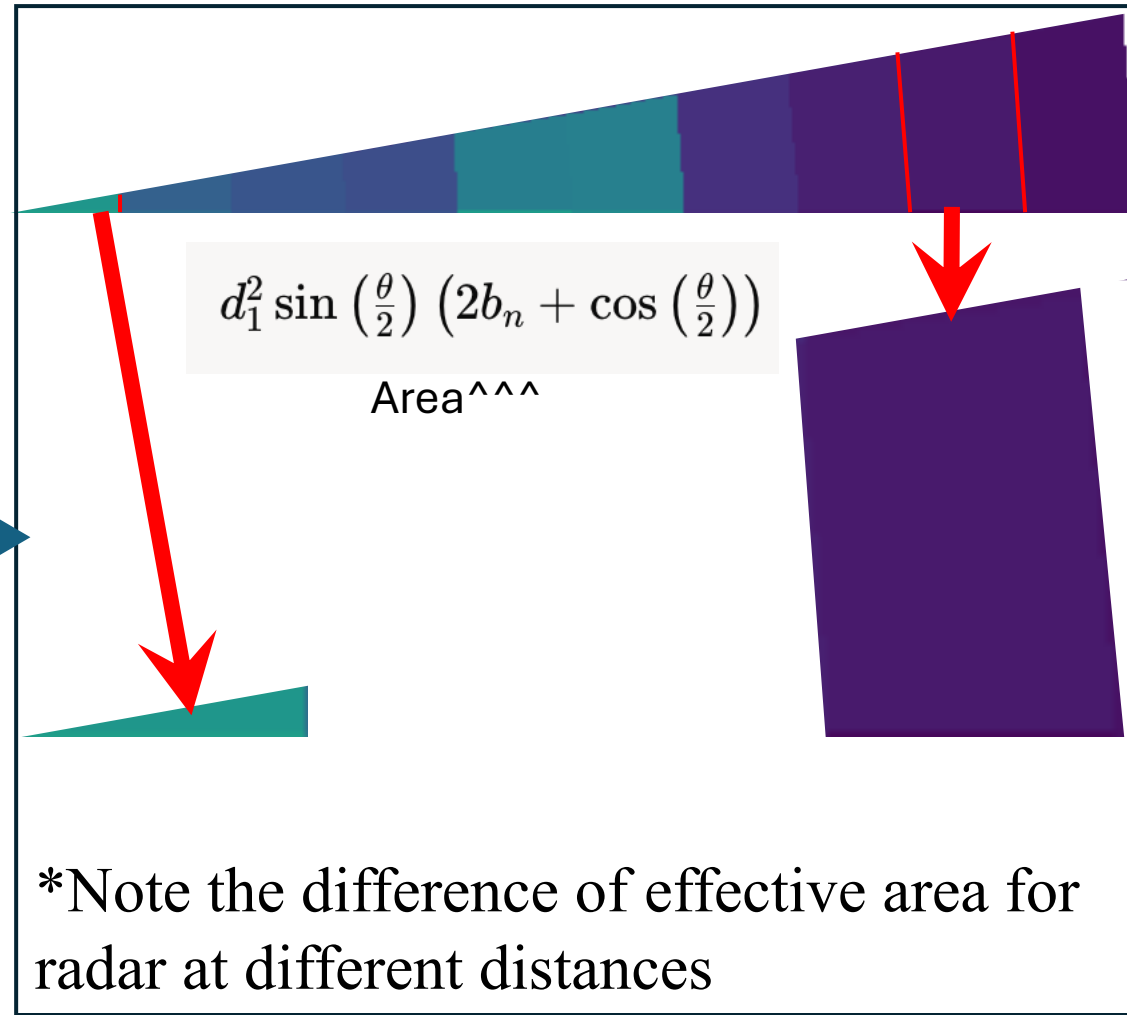
Radar Simulations - Explainer

Simplified model – 10 bins, 36 rays, 4 bit



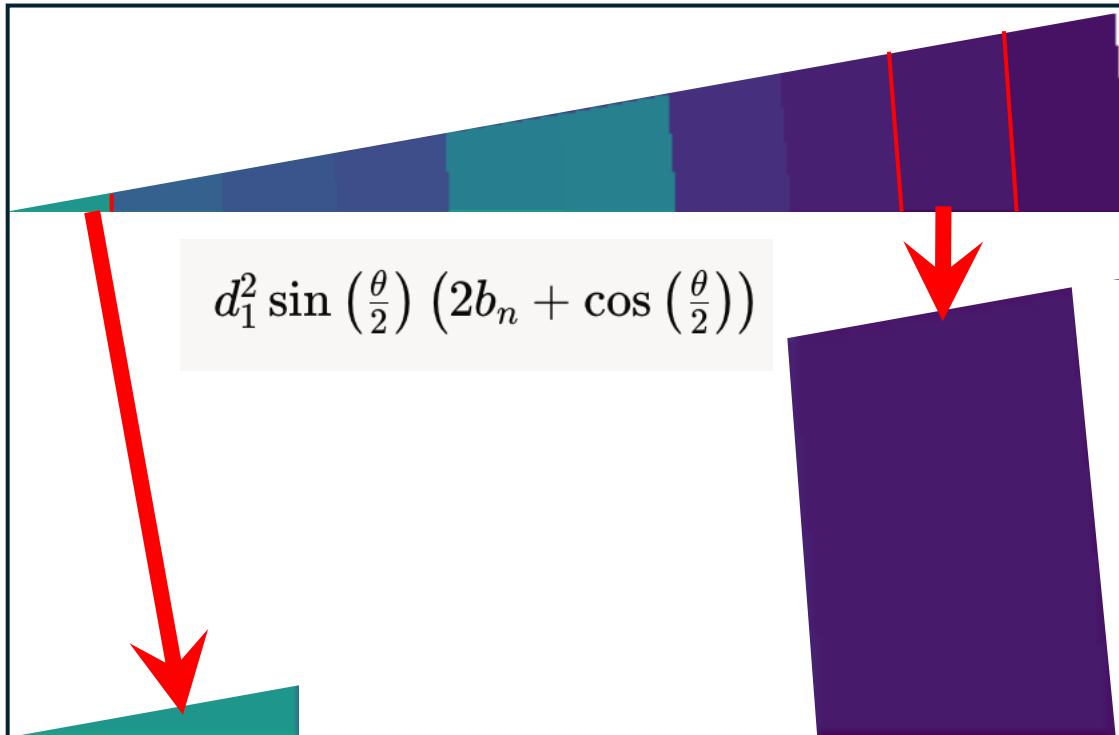
SIMPLIFY

Simple Simulation – Single Ray



Radar Simulations - Explainer

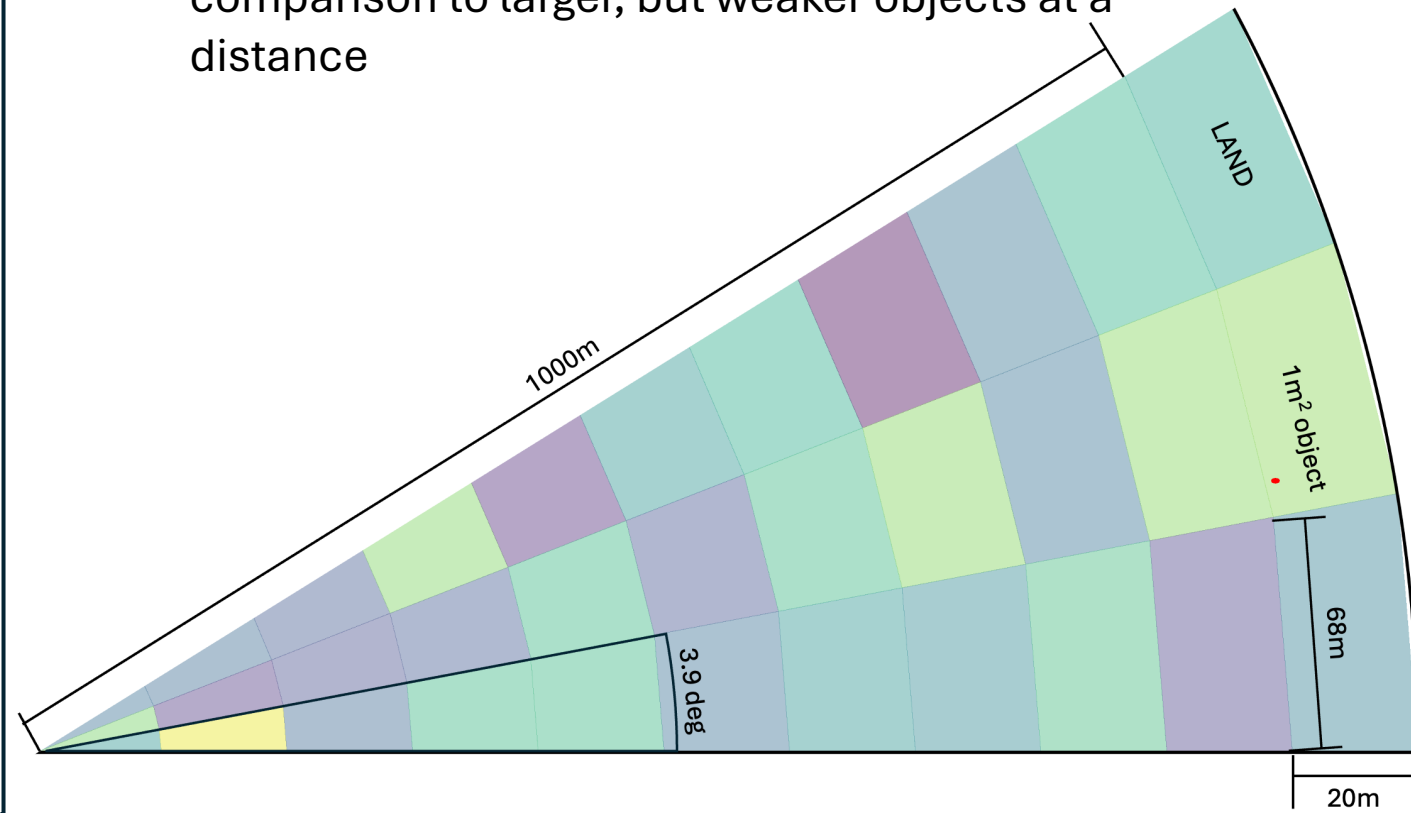
Simple Simulation – Single Ray



*Note the difference of effective area for radar at different distances

Each area can be quite large, $\sim 1372\text{m}^2$ at 1km
And only 13.6m^2 at the radar.

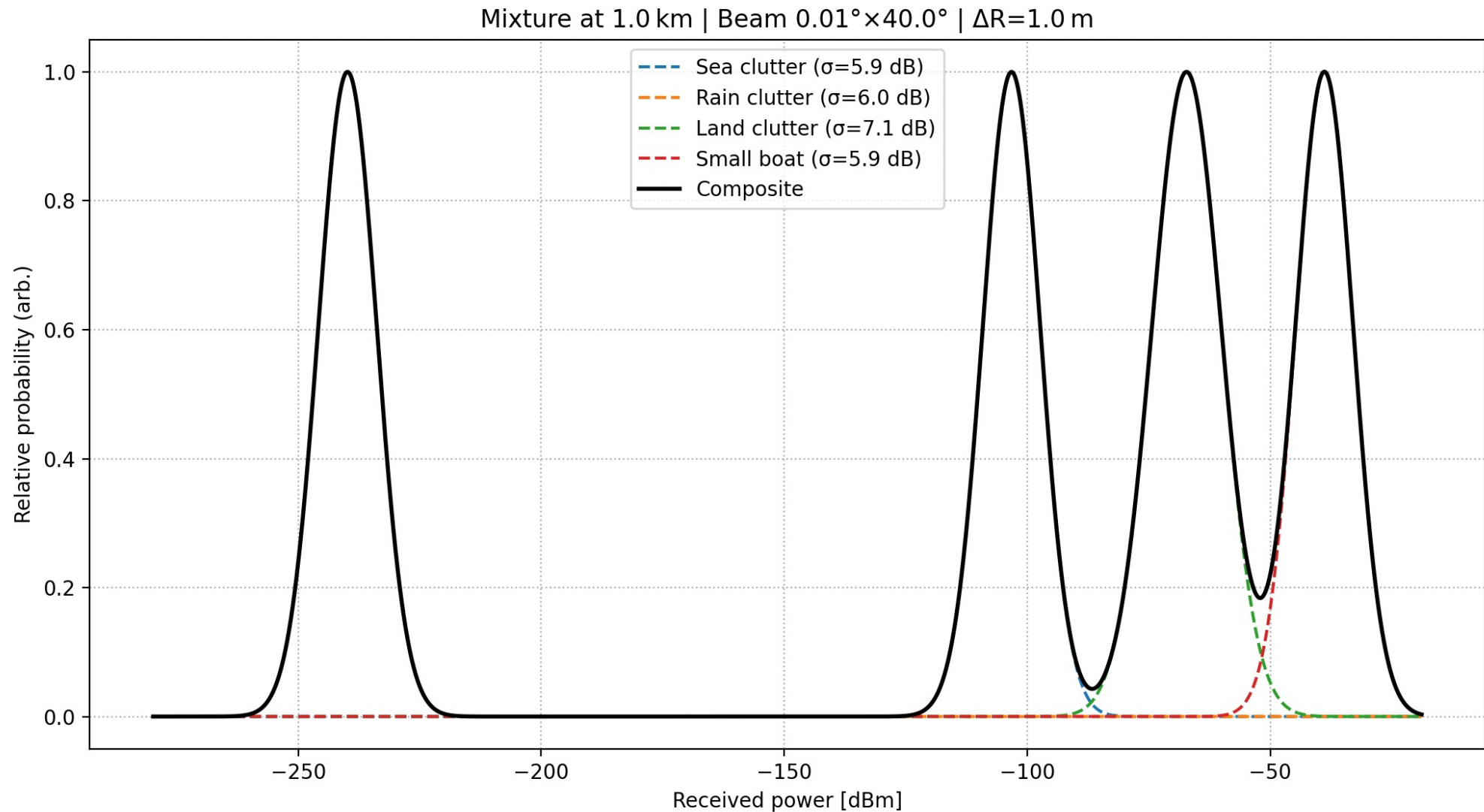
Buoys and other small objects get harder to detect in comparison to larger, but weaker objects at a distance



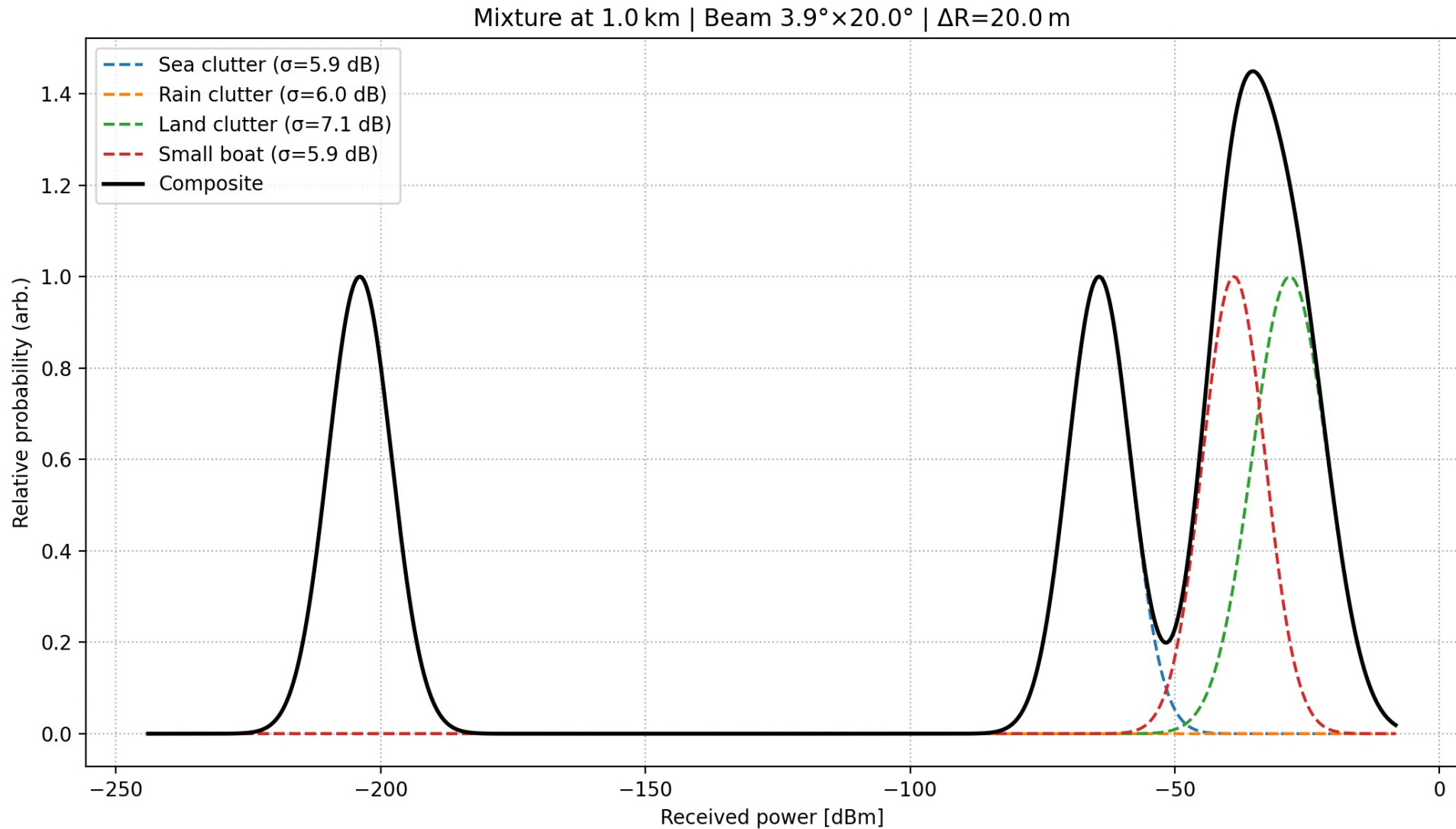
Radar Simulations - Explainer

- So, how do we determine desired objects vs random reflections(i.e. sea) at that distance?
- Requires understanding the radar(i.e. characteristics) AND understanding the makeup of the objects within each radial bin and their radar properties (RCS, reflectivity, etc)
- Therefore, to develop a simulation, there needs to be a per-bin model of returns.

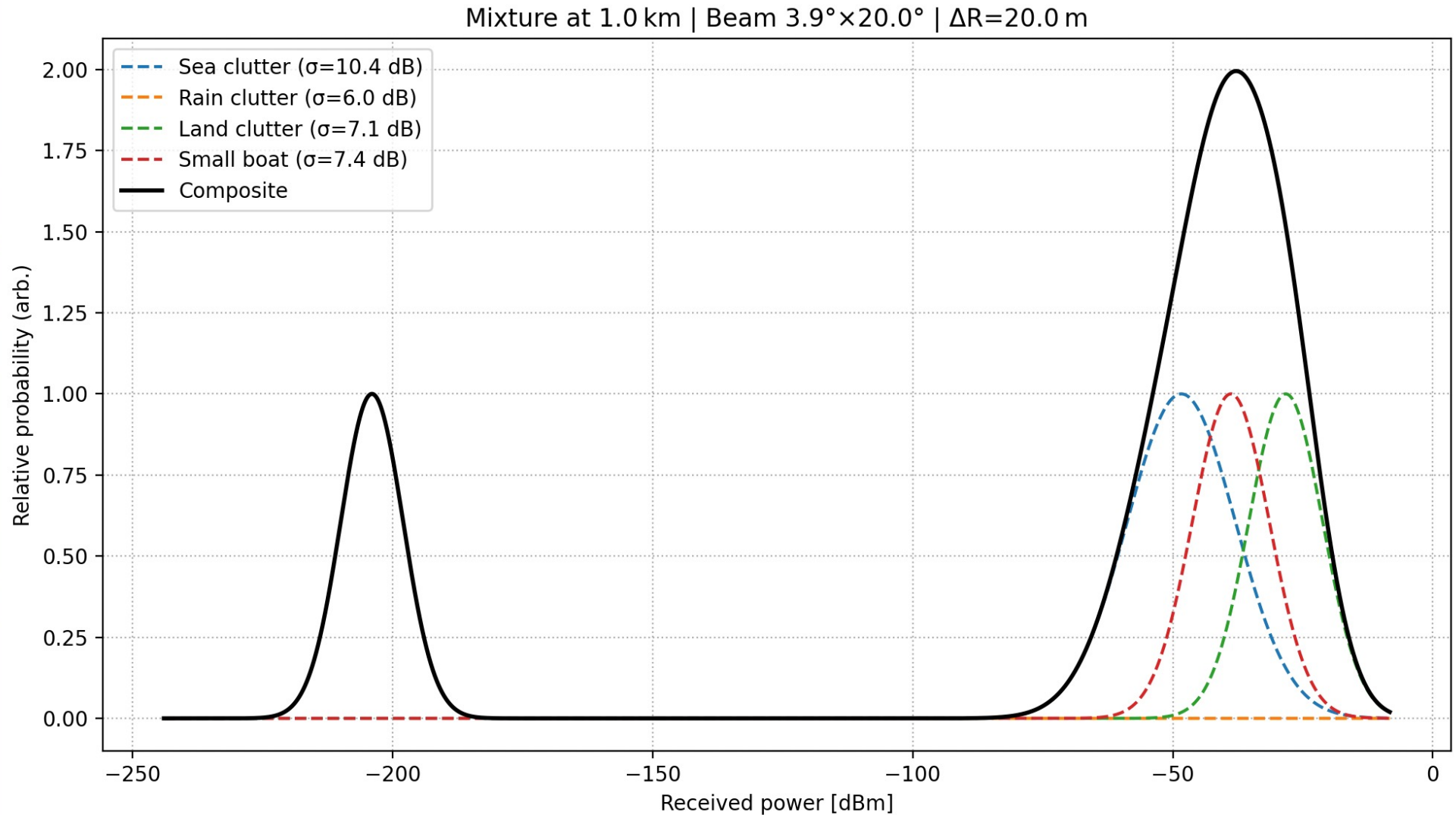
Clear separation



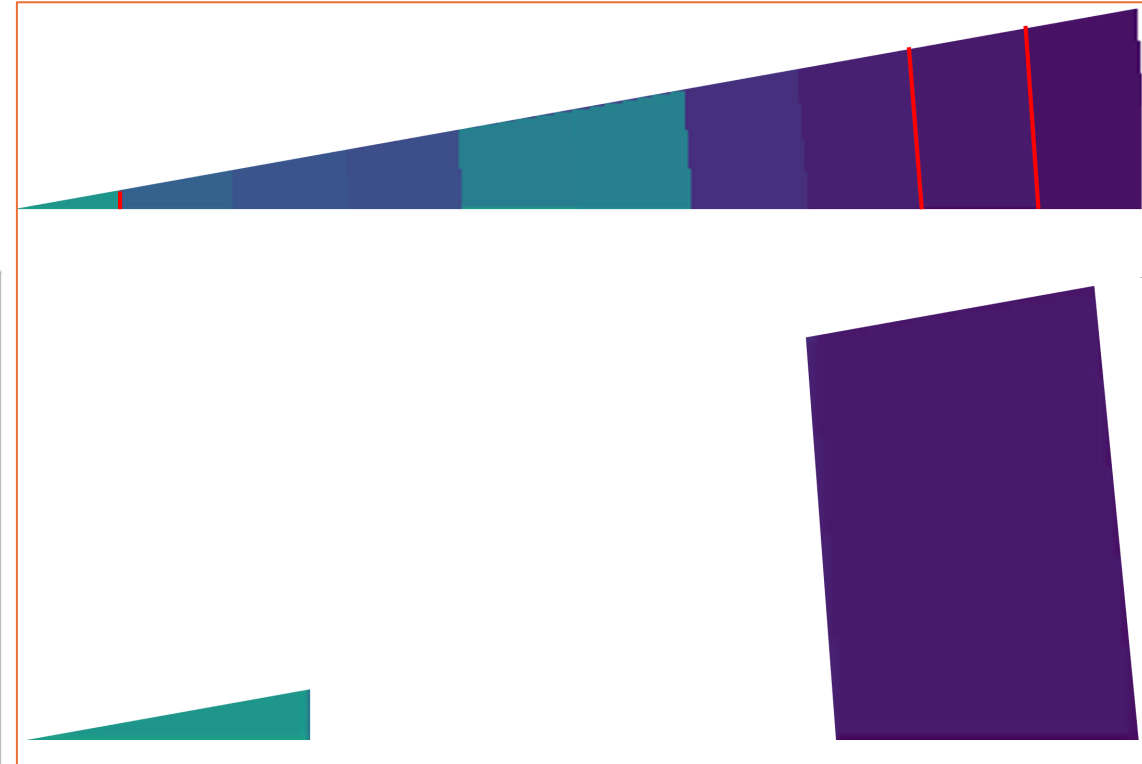
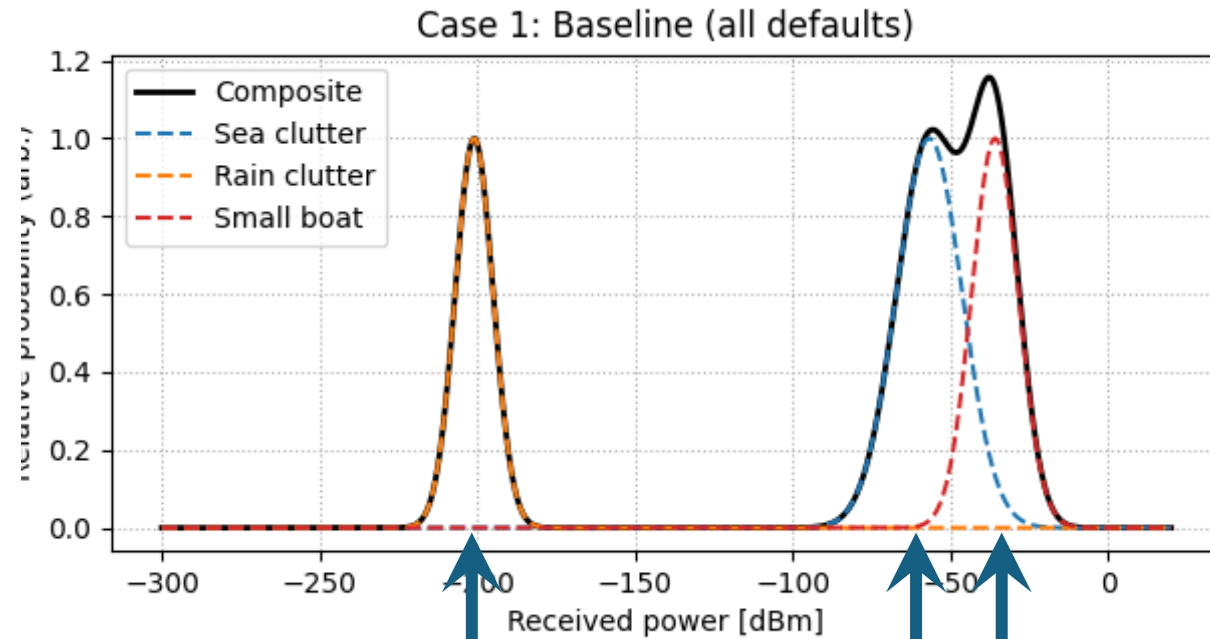
Data with real radar params



High sea state



Algorithm



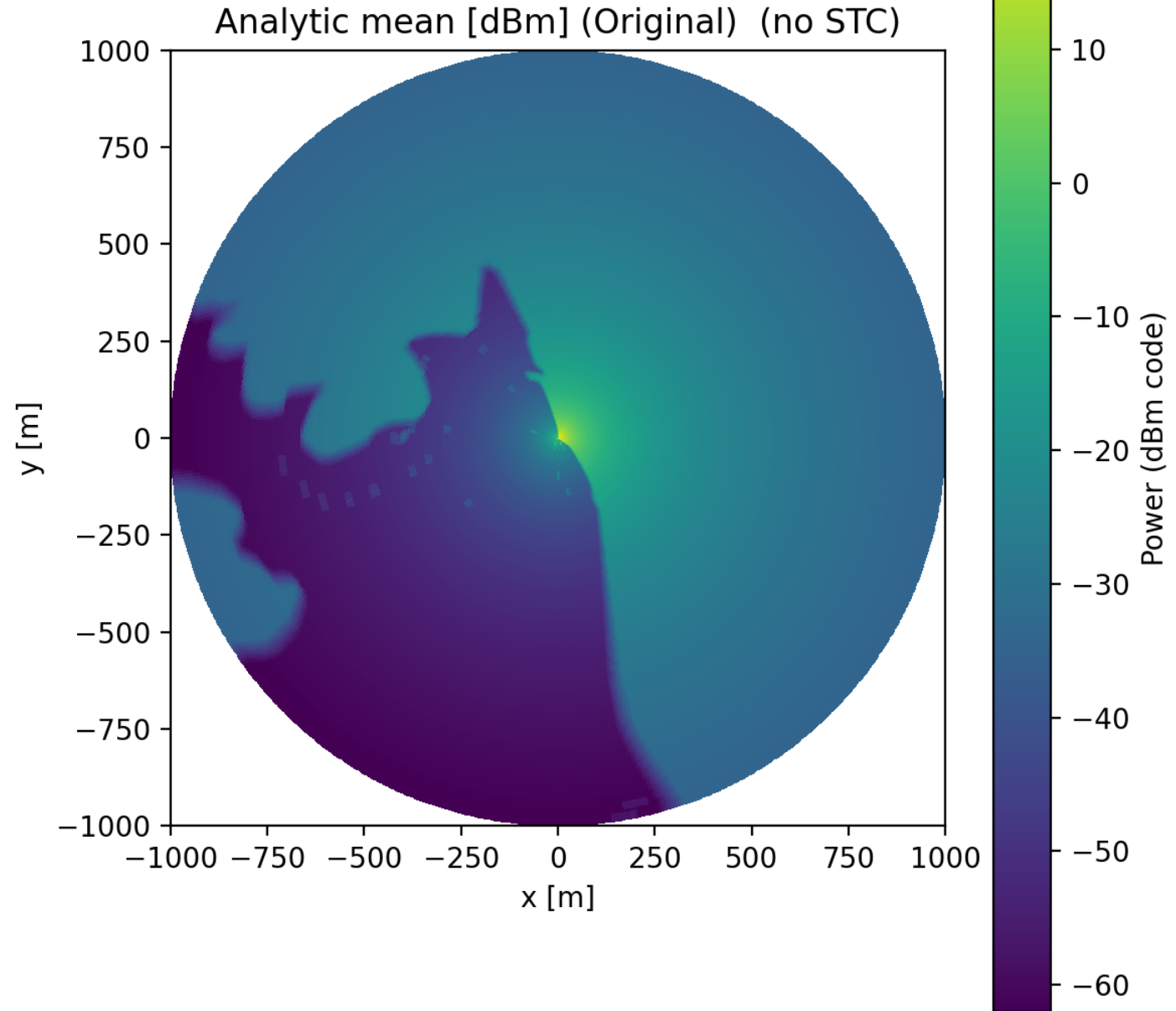
Random selection of values from distribution

Values summed to create final 'return signal'

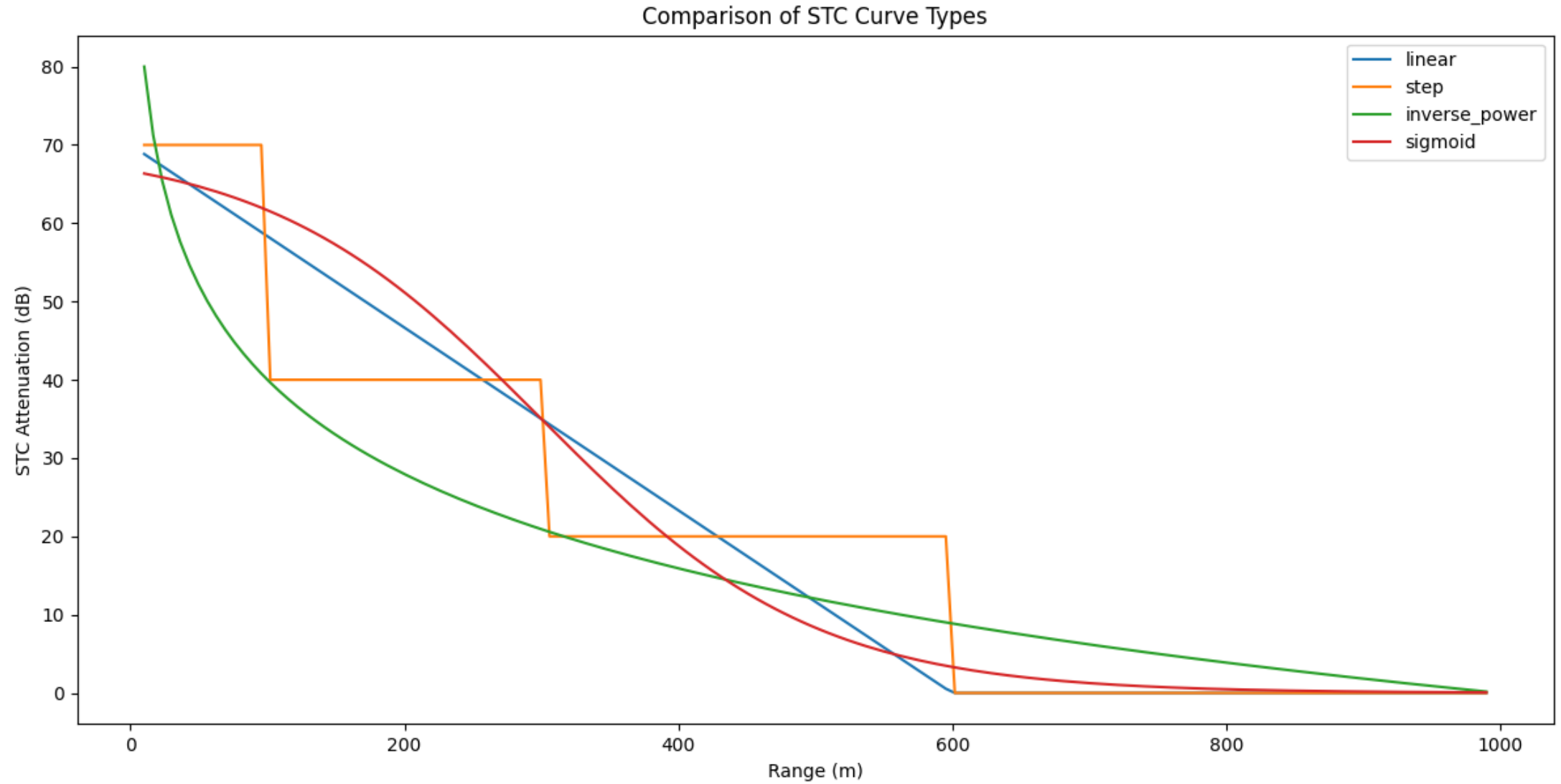
Raw output:

The center of the radar is much stronger than the edges (range⁴ rule)
So, most radars apply some type of 'STC Curve' or Sensitivity Time Control.

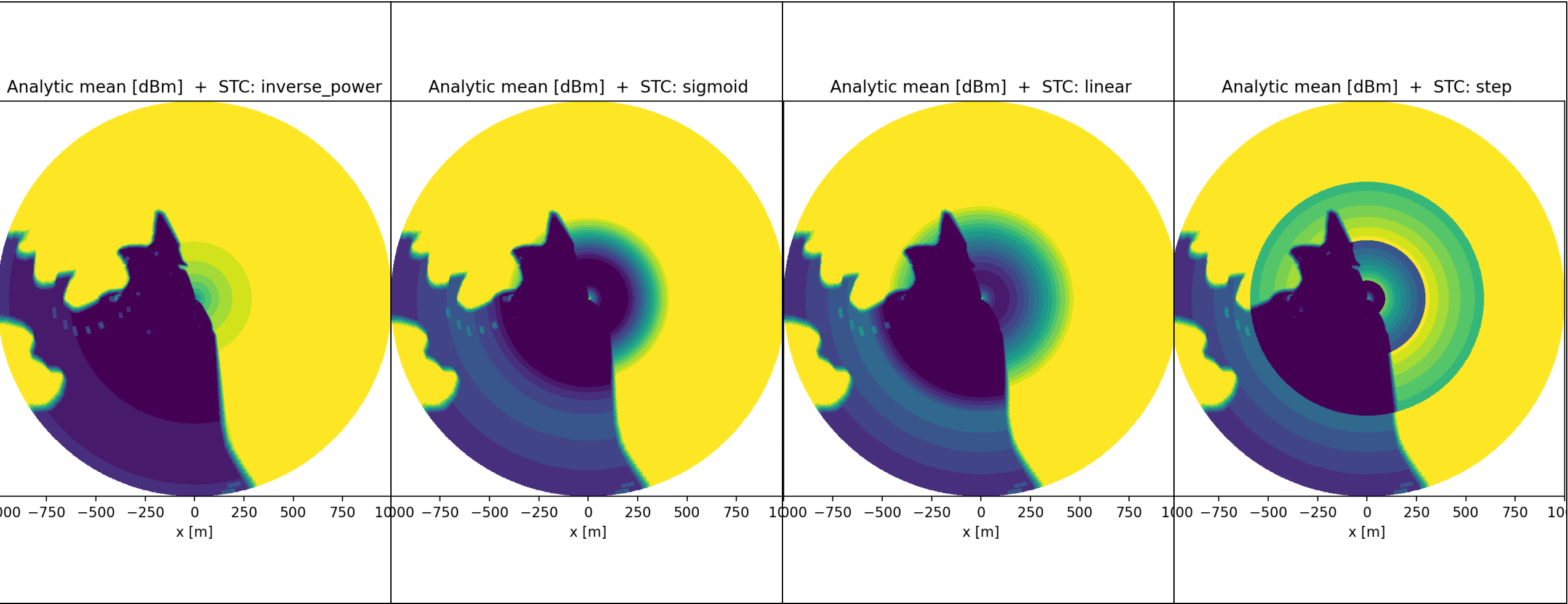
These can be a variety of shapes, but they all work similarly by attenuating signals closer to the radar and slowly reducing this attenuation as range increases. It is generally time based because that was easier to implement on analog circuitry.



Different STC Curves

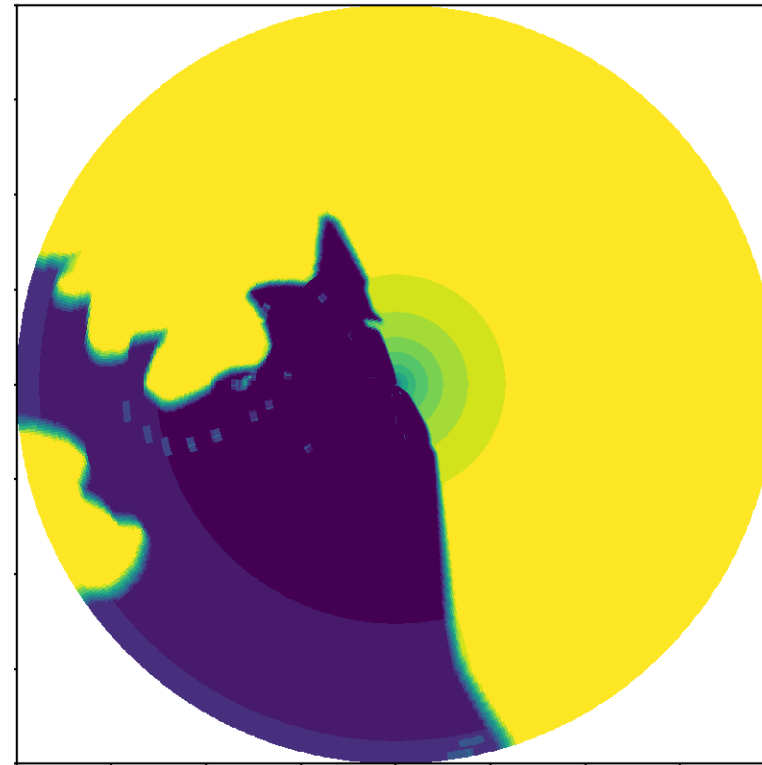


STC results

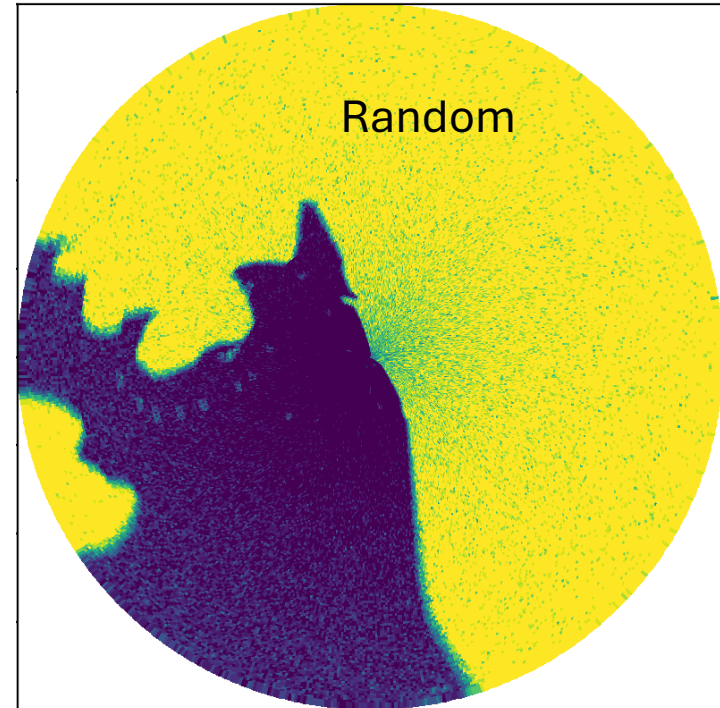


Different random selections

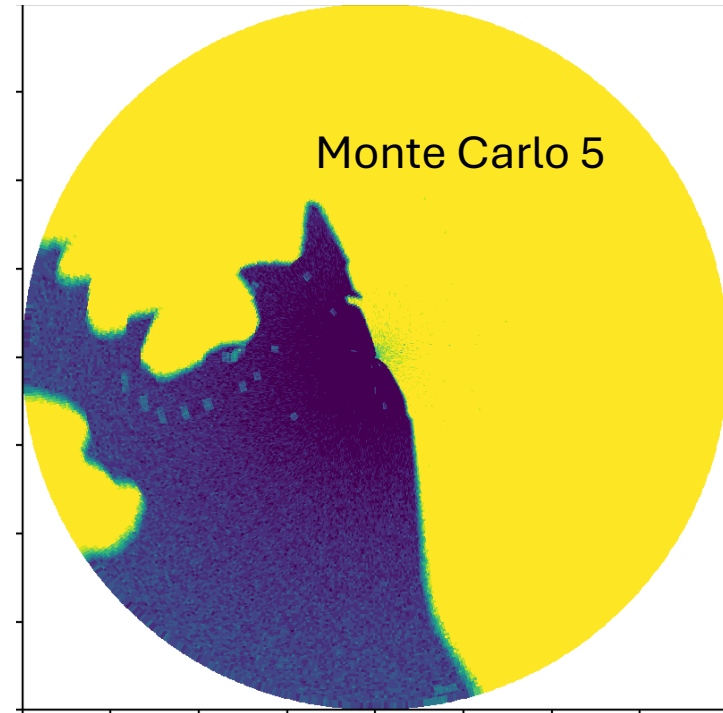
- For the gaussian mix for each bin, random selection of a value based on the gaussian curve works, but so does using the analytical mean.
- Both provide some value, but create slightly artificial and non-modifiable outputs.
- Therefore, a Monte Carlo simulation version is included to allow for finer control
- *note results on this slide are pre-quantized and have STC r^2 law applied



Analytic



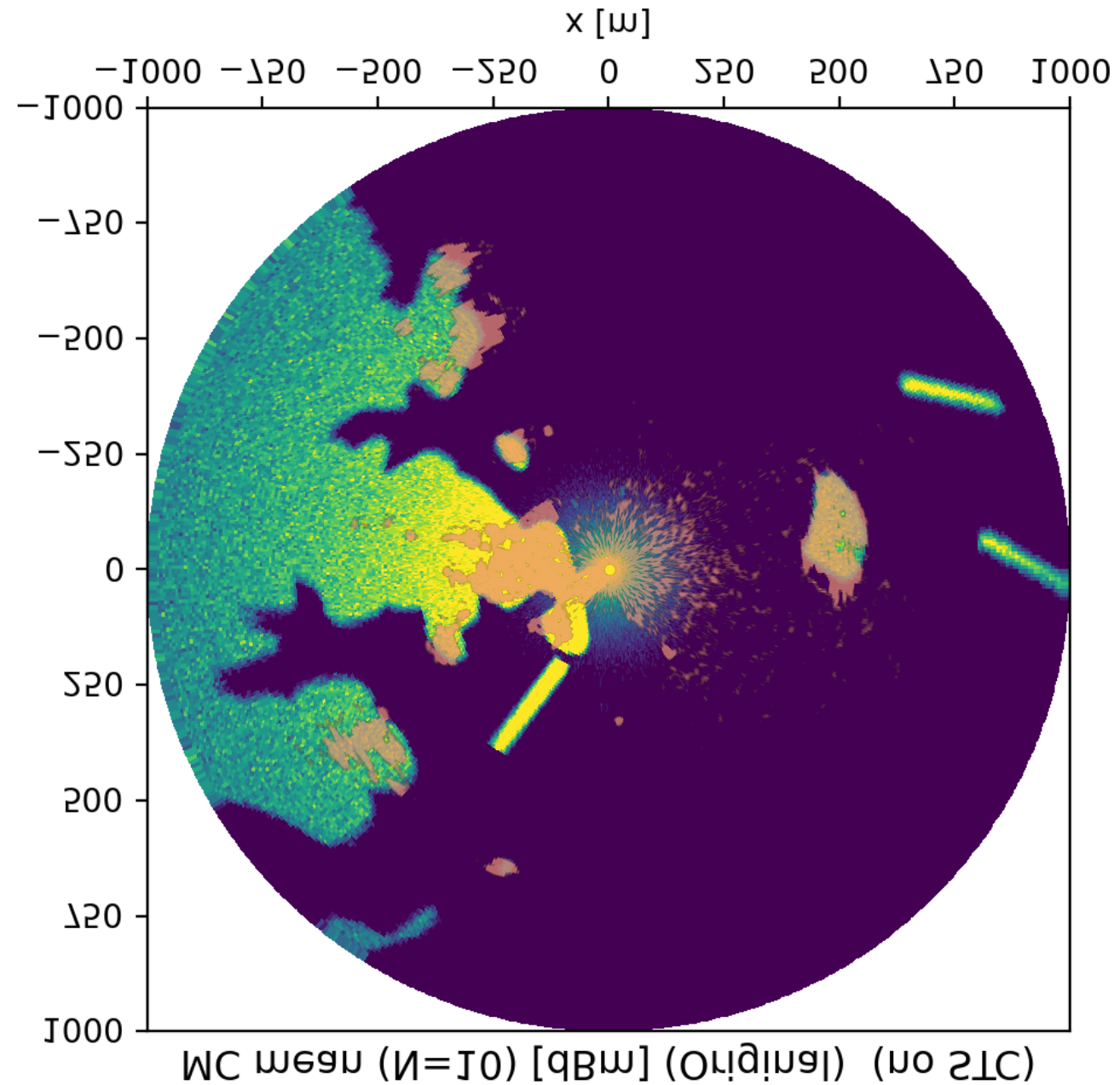
Random



Monte Carlo 5

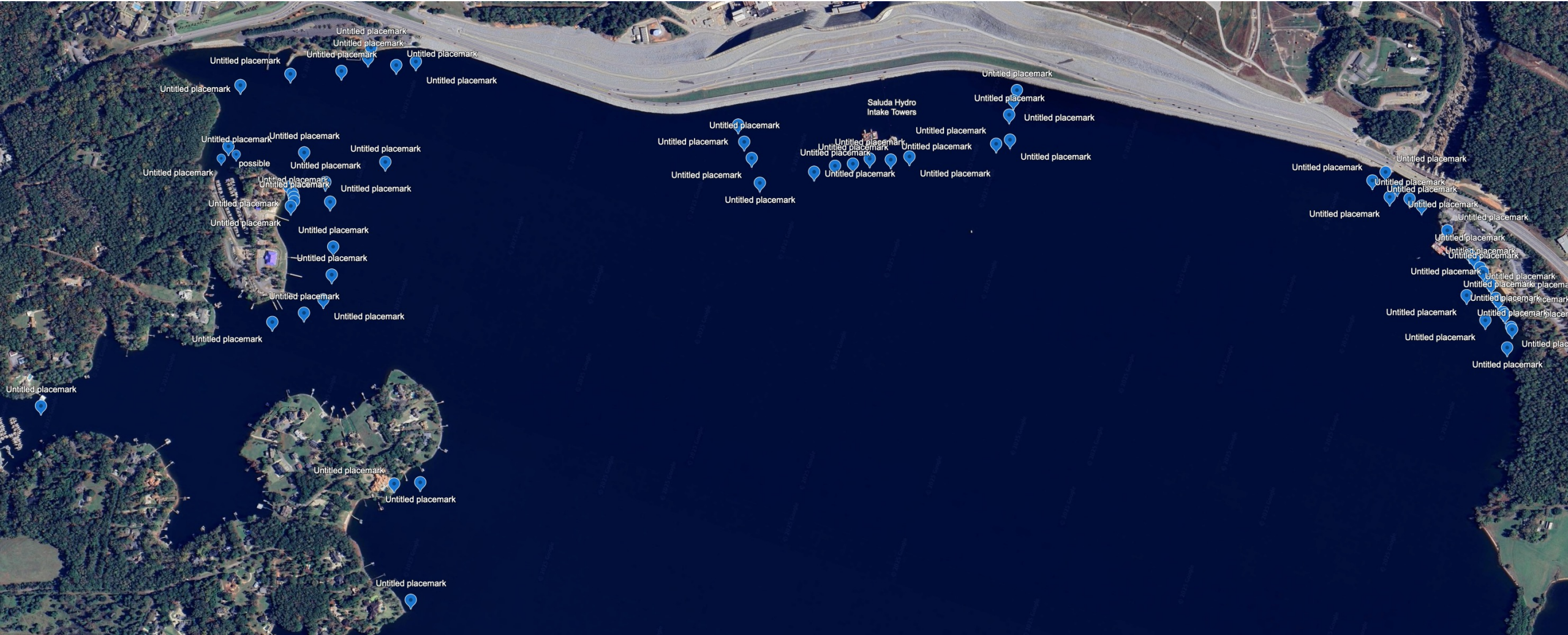
Comparing to real data

- Data from Lutheran church @ lake murray



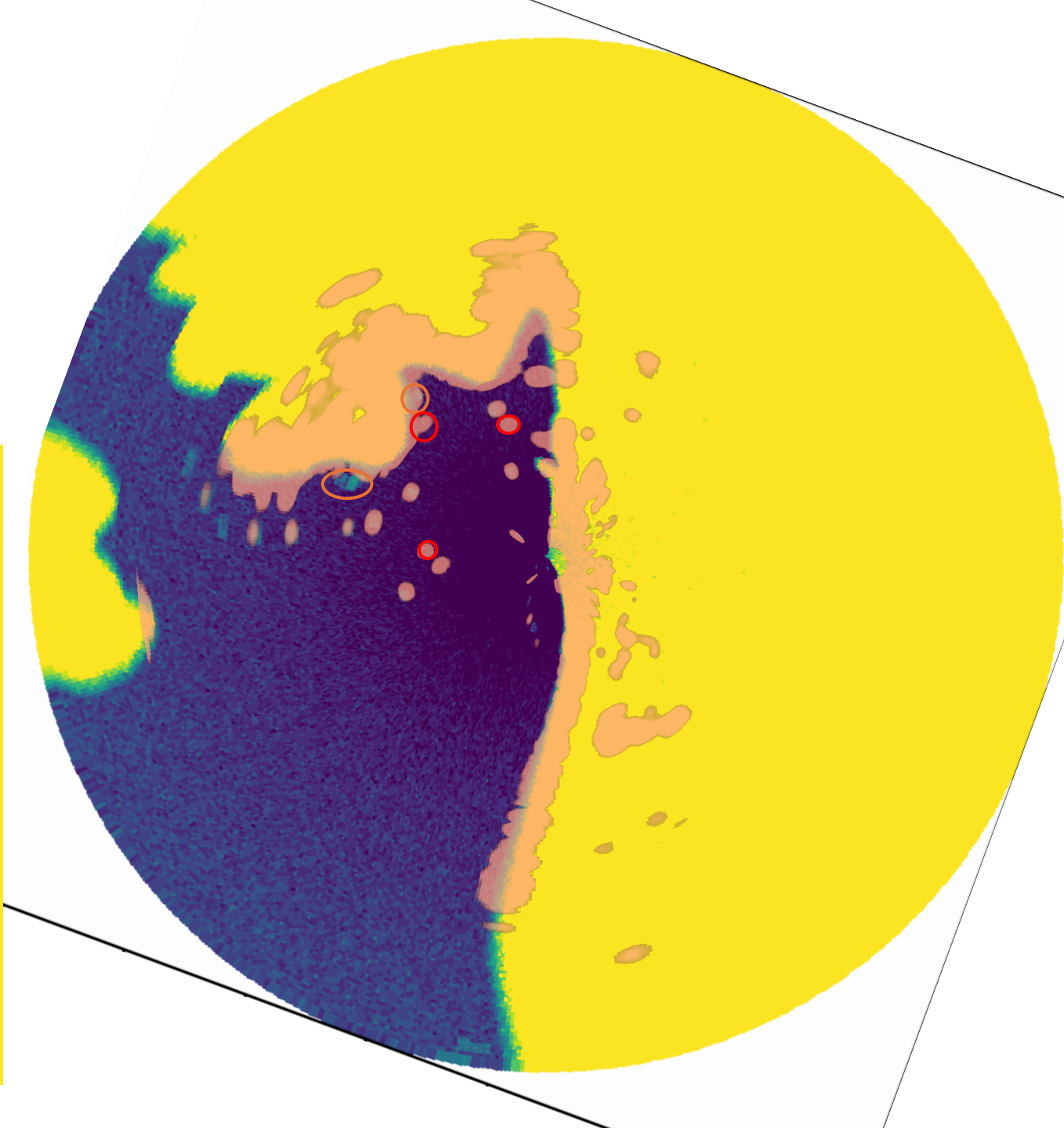
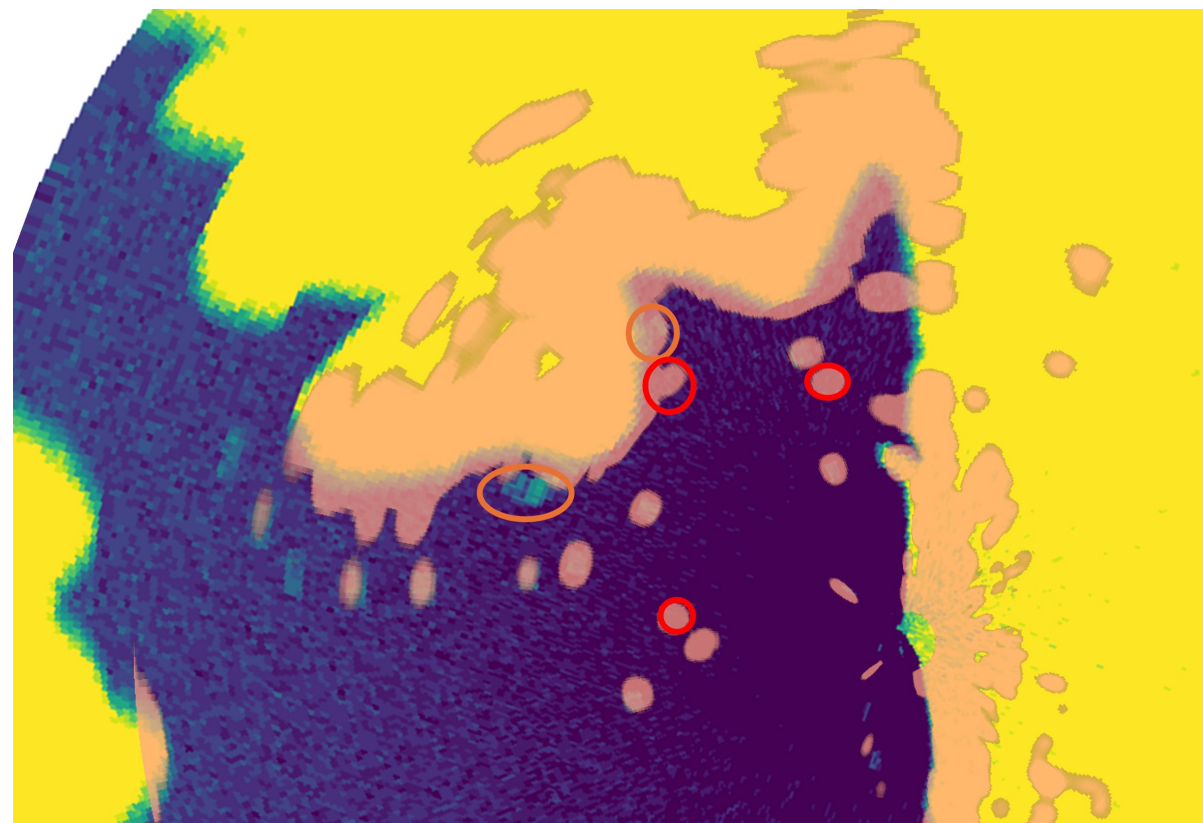
Added Realism

- Added buoys and land masses from GPS coords. (Nov 13, '24)



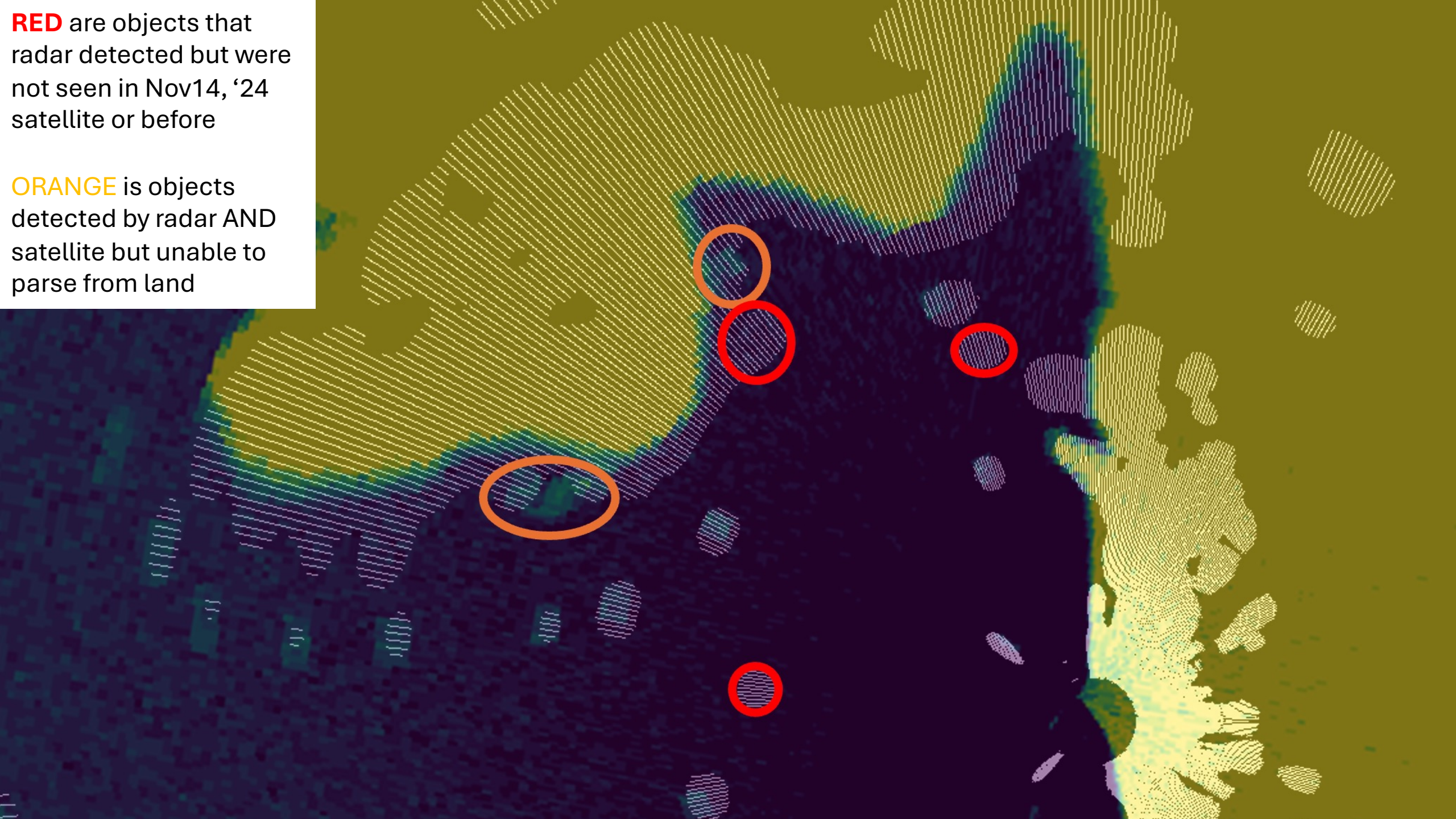
Simulation vs Radar results

- Data from Murray dam



RED are objects that
radar detected but were
not seen in Nov14, '24
satellite or before

ORANGE is objects
detected by radar AND
satellite but unable to
parse from land



Characterization Test #1

reflector 1



reflector 2



radar



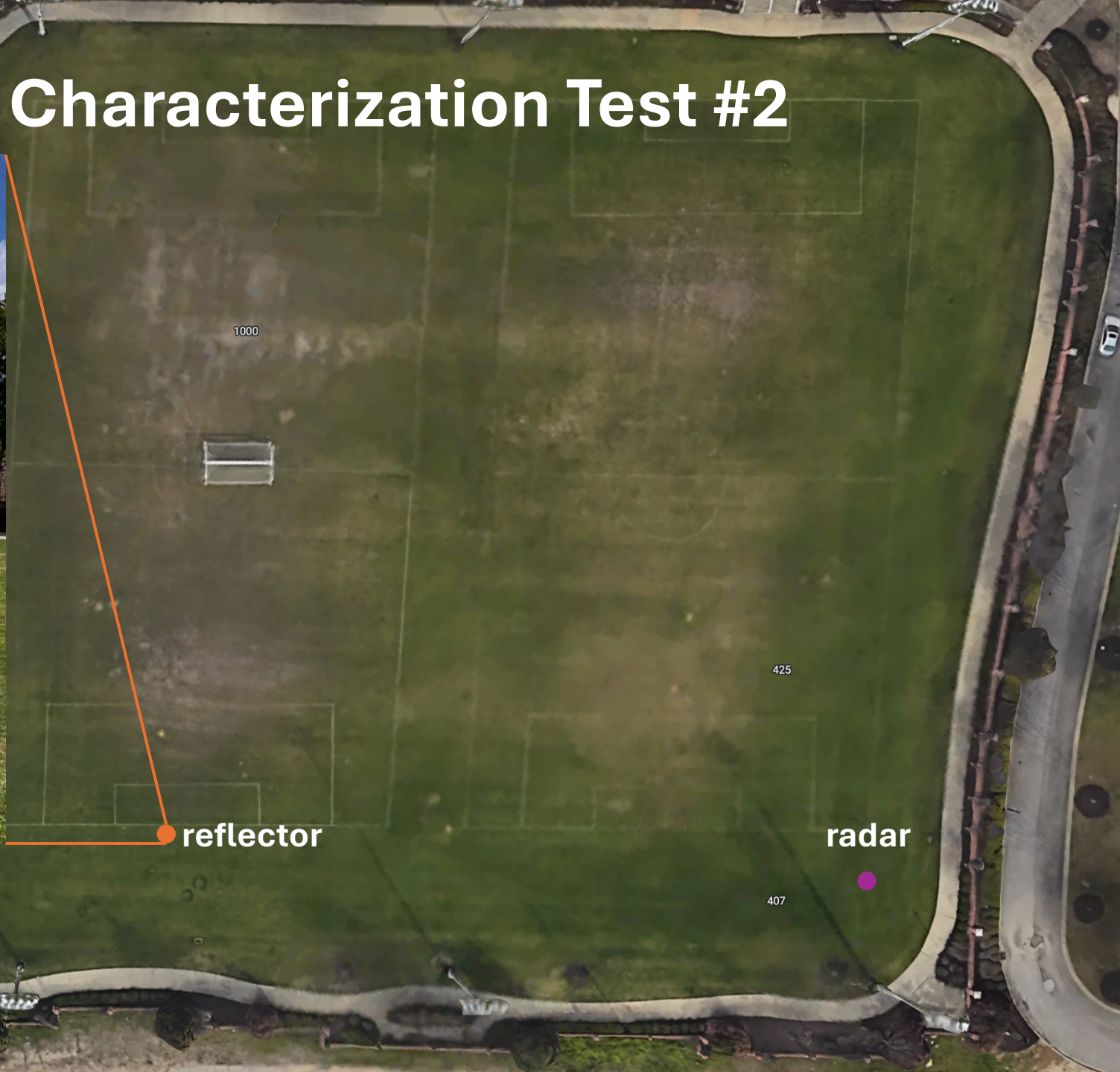
400

1000

425

407

Characterization Test #2





reflector

radar 407

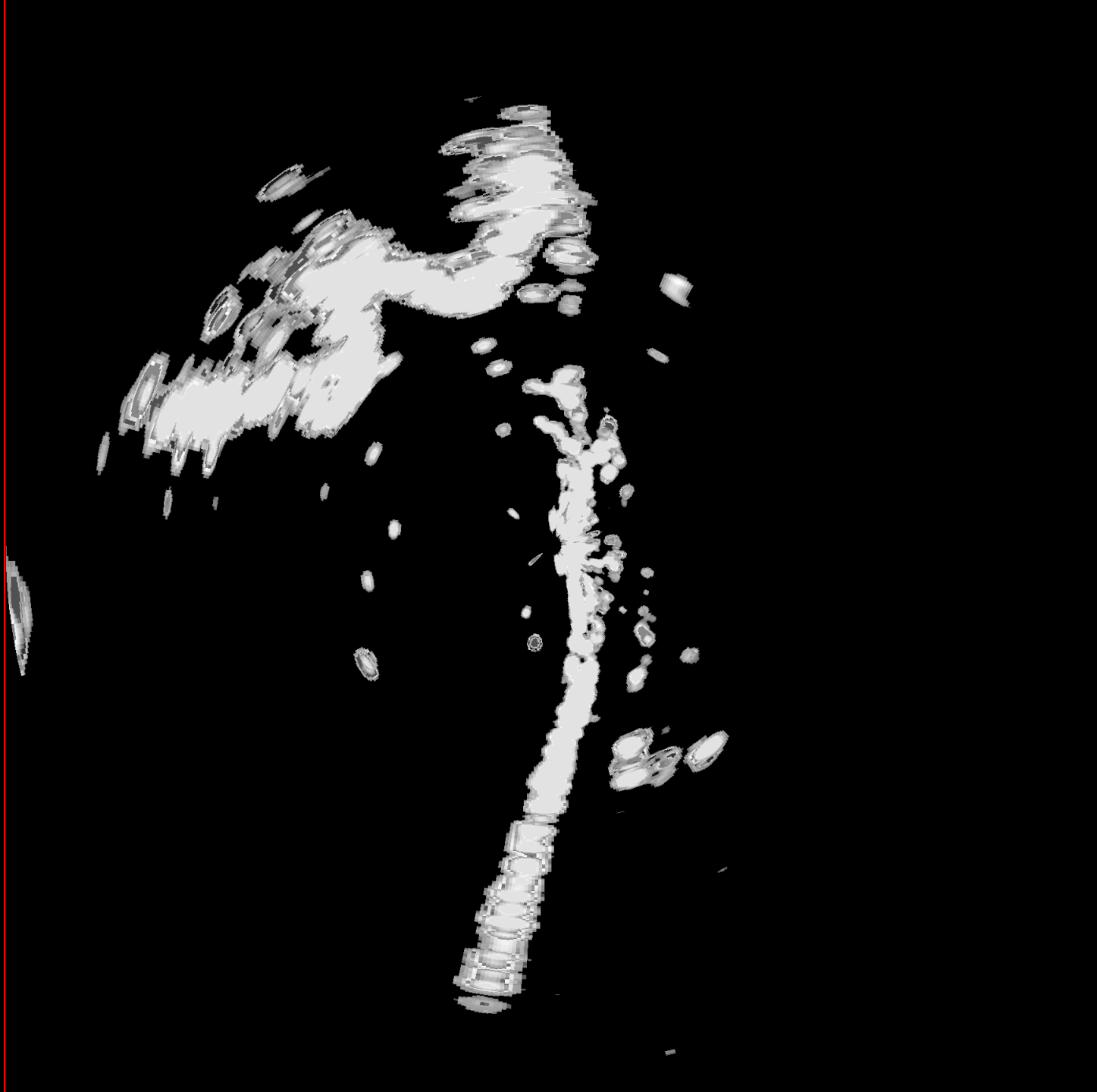
Object tracking – blob separation



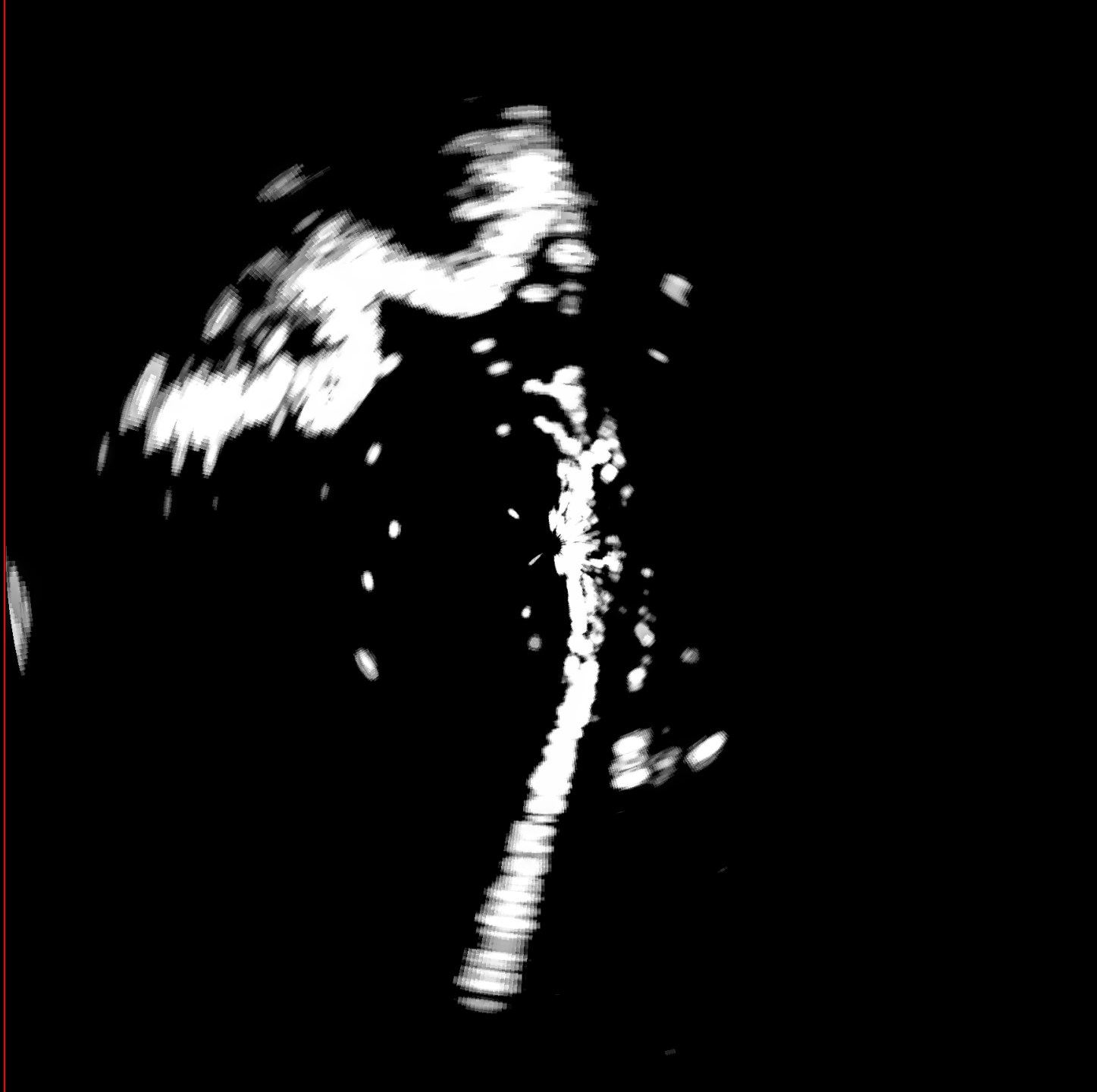
Simply stacking
transparent
frames (SSTF)



Debevec
OpenCV



Merge Mertens OpenCV



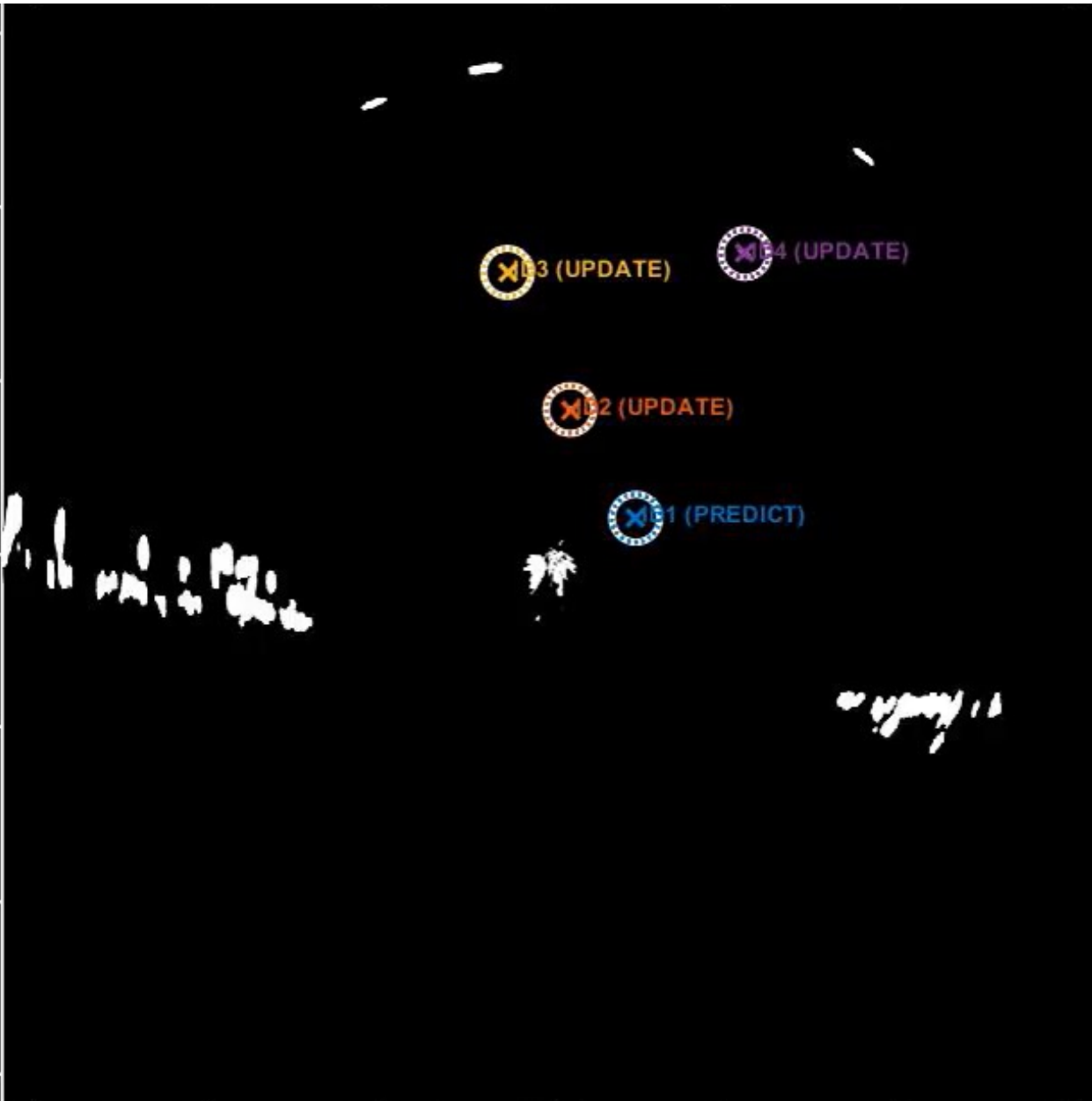
Robertson

OpenCV



Ty; Charleston Data

Auto tracking implementation



Murray Data; Low sensitivity

Auto tracking implementation



Murray Data; High sensitivity

Auto tracking implementation



Murray Data; Kalman - lite

Auto tracking implementation

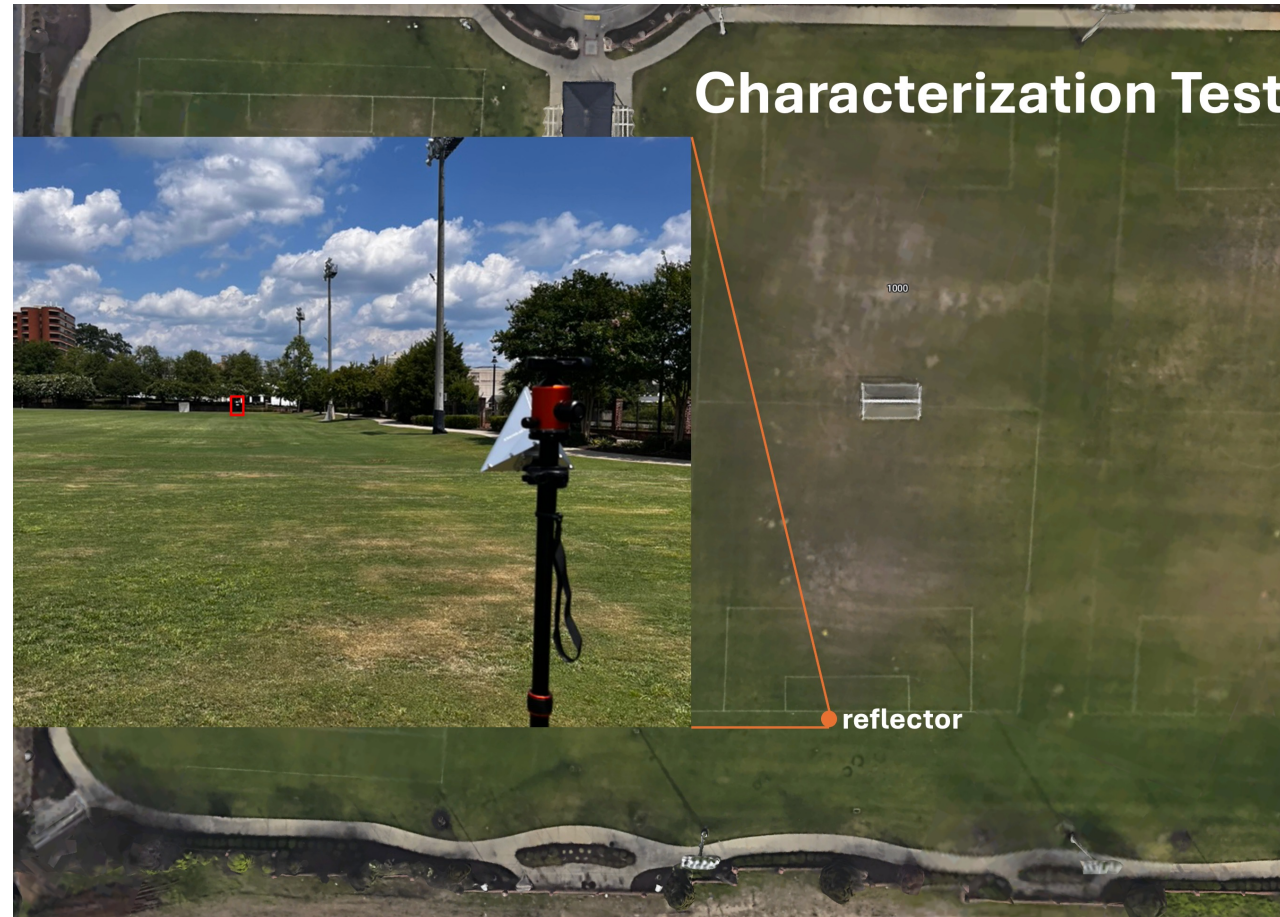
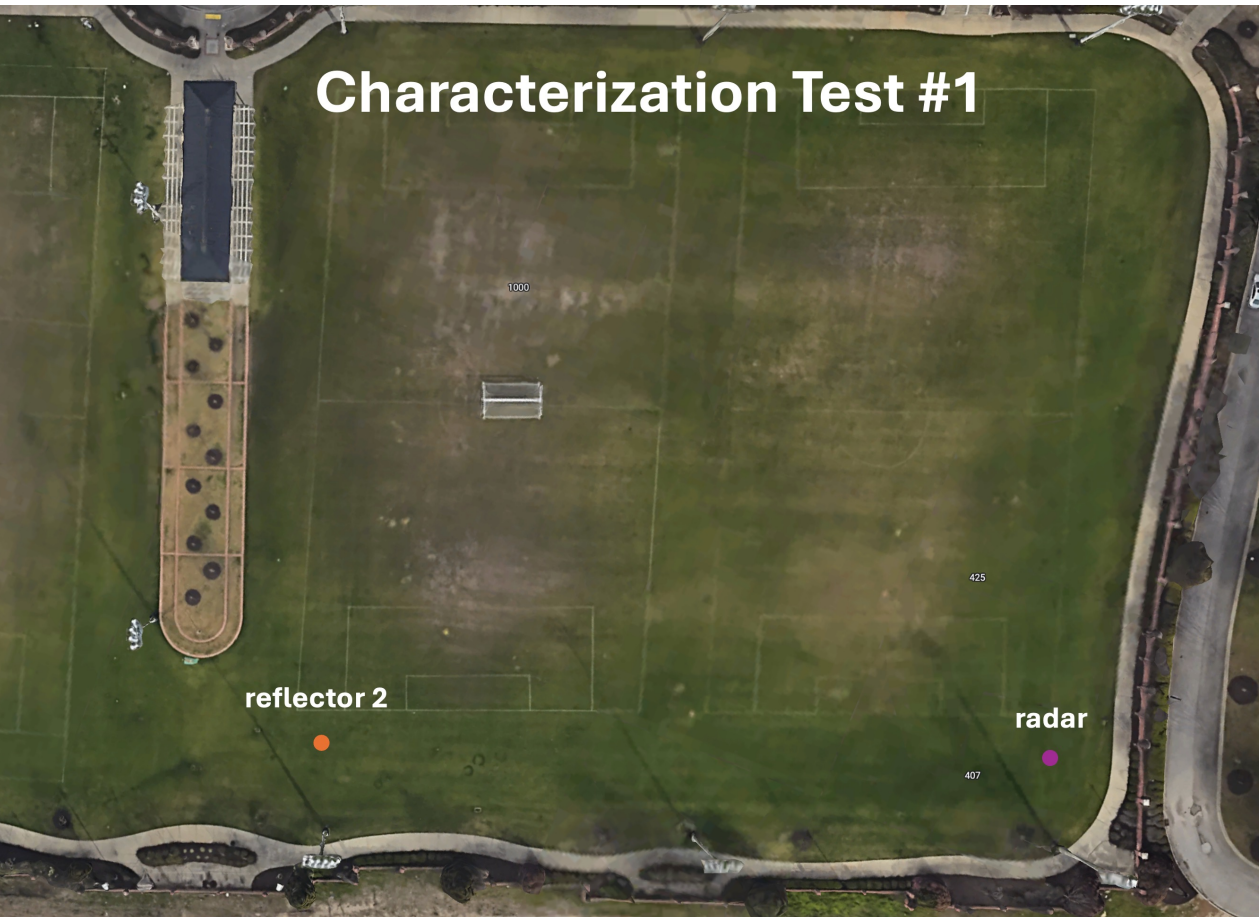
Frame 0

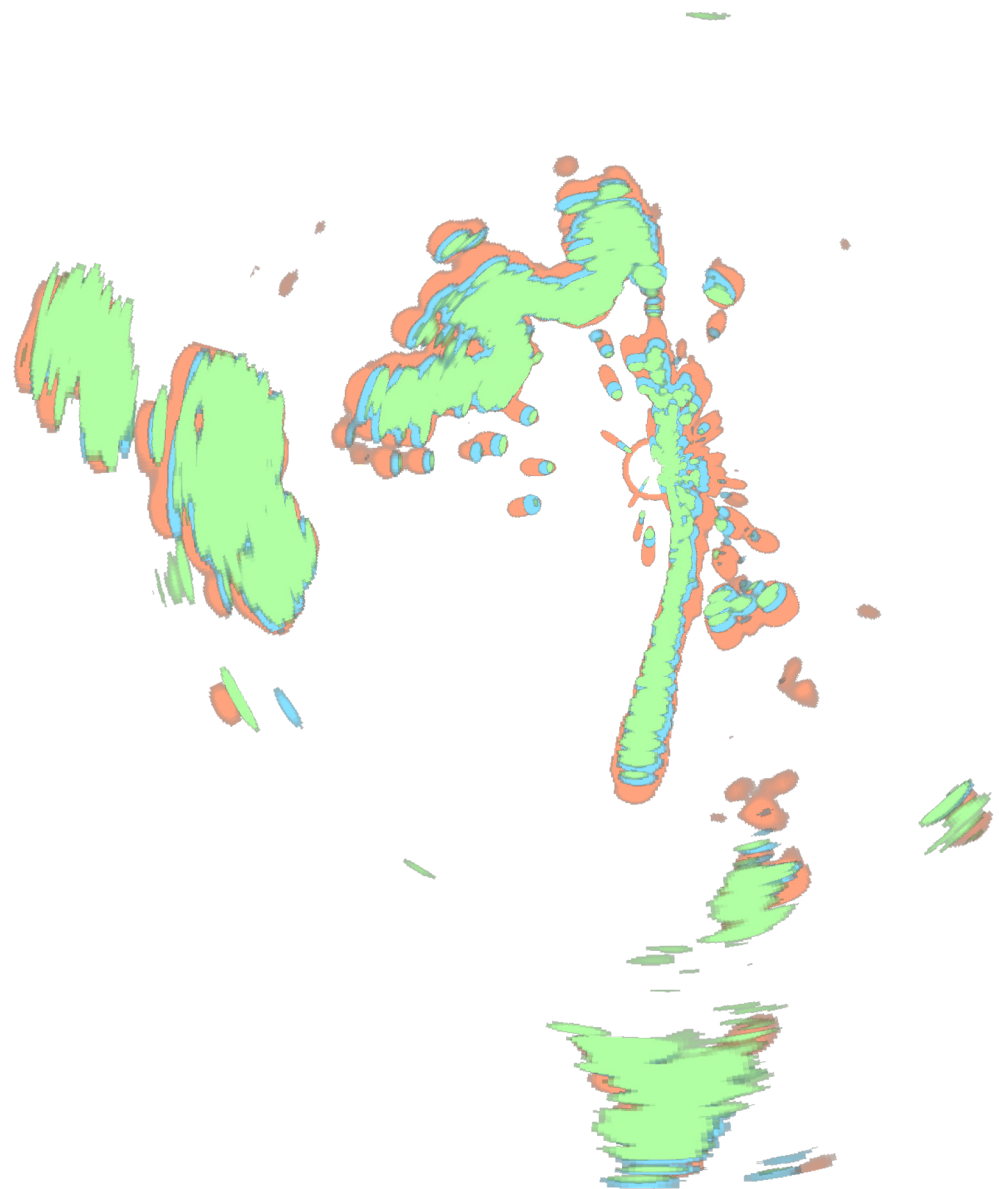


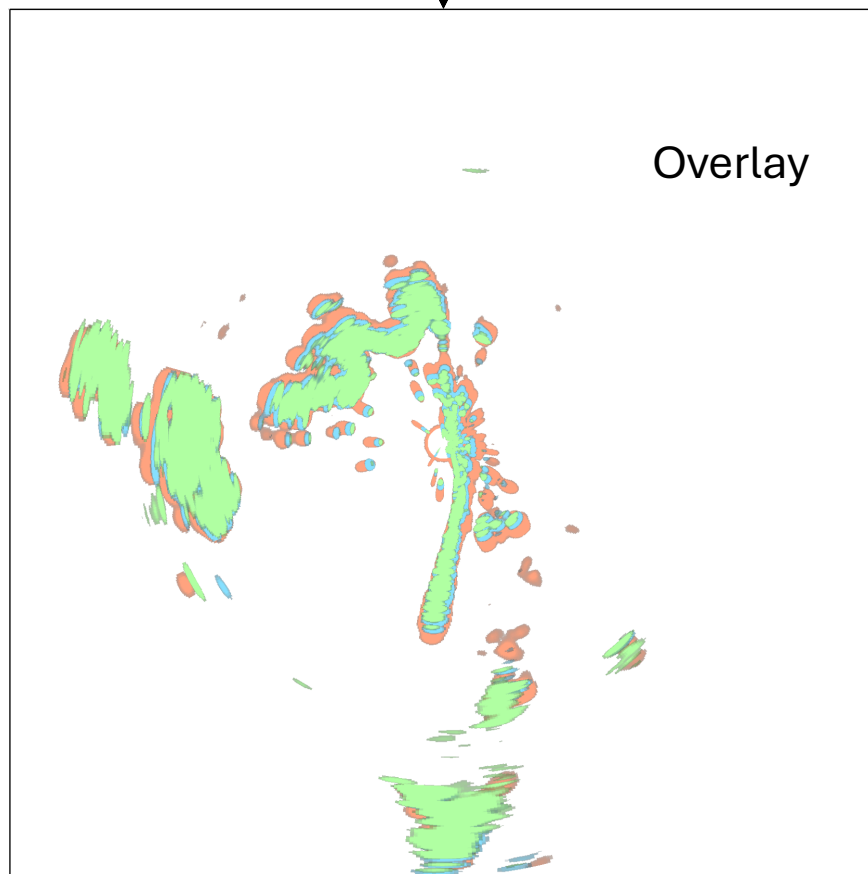
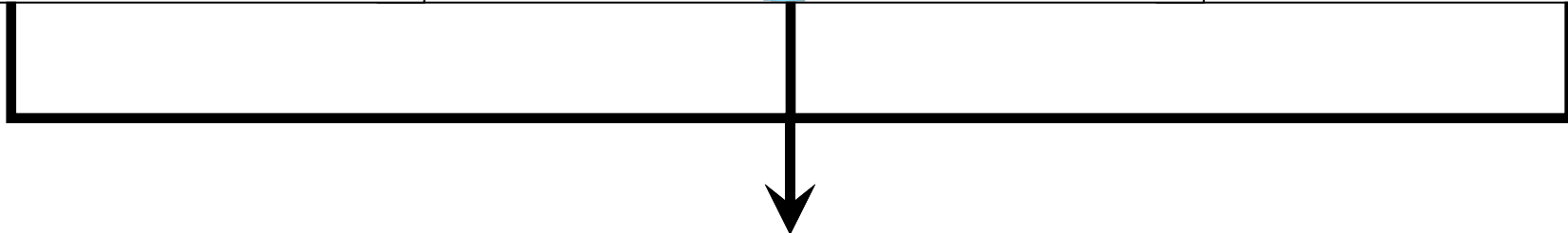
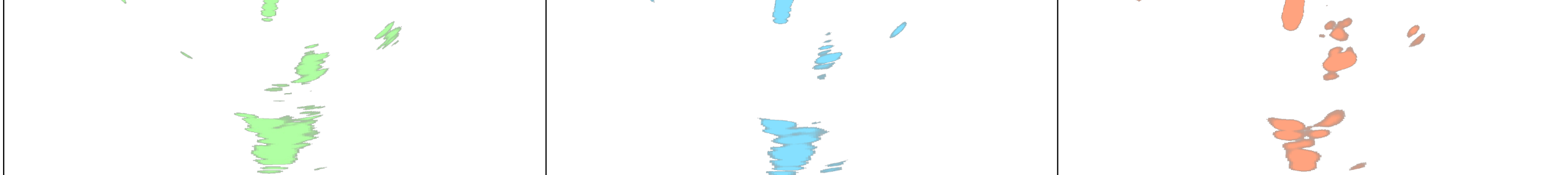
Future work

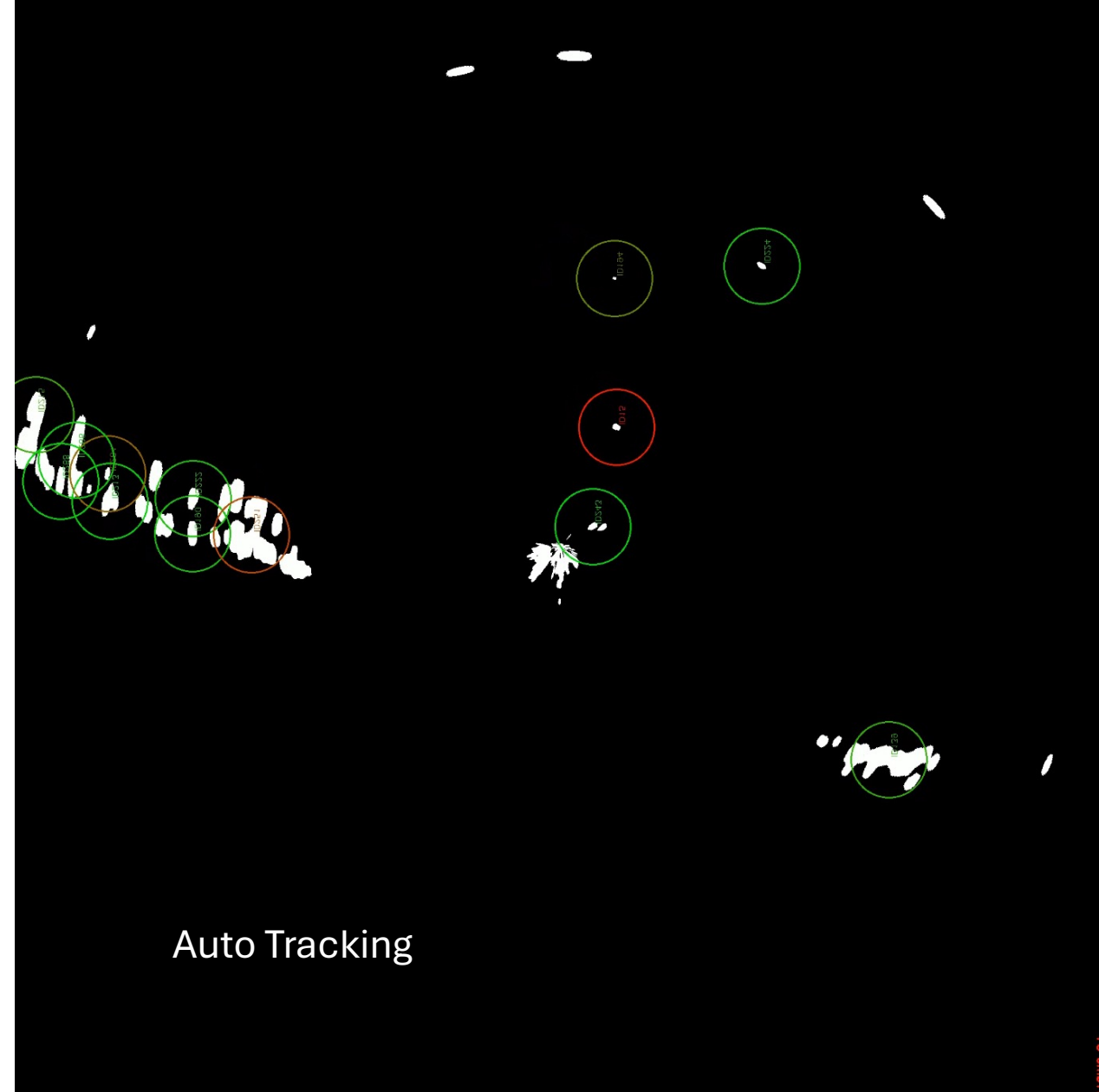
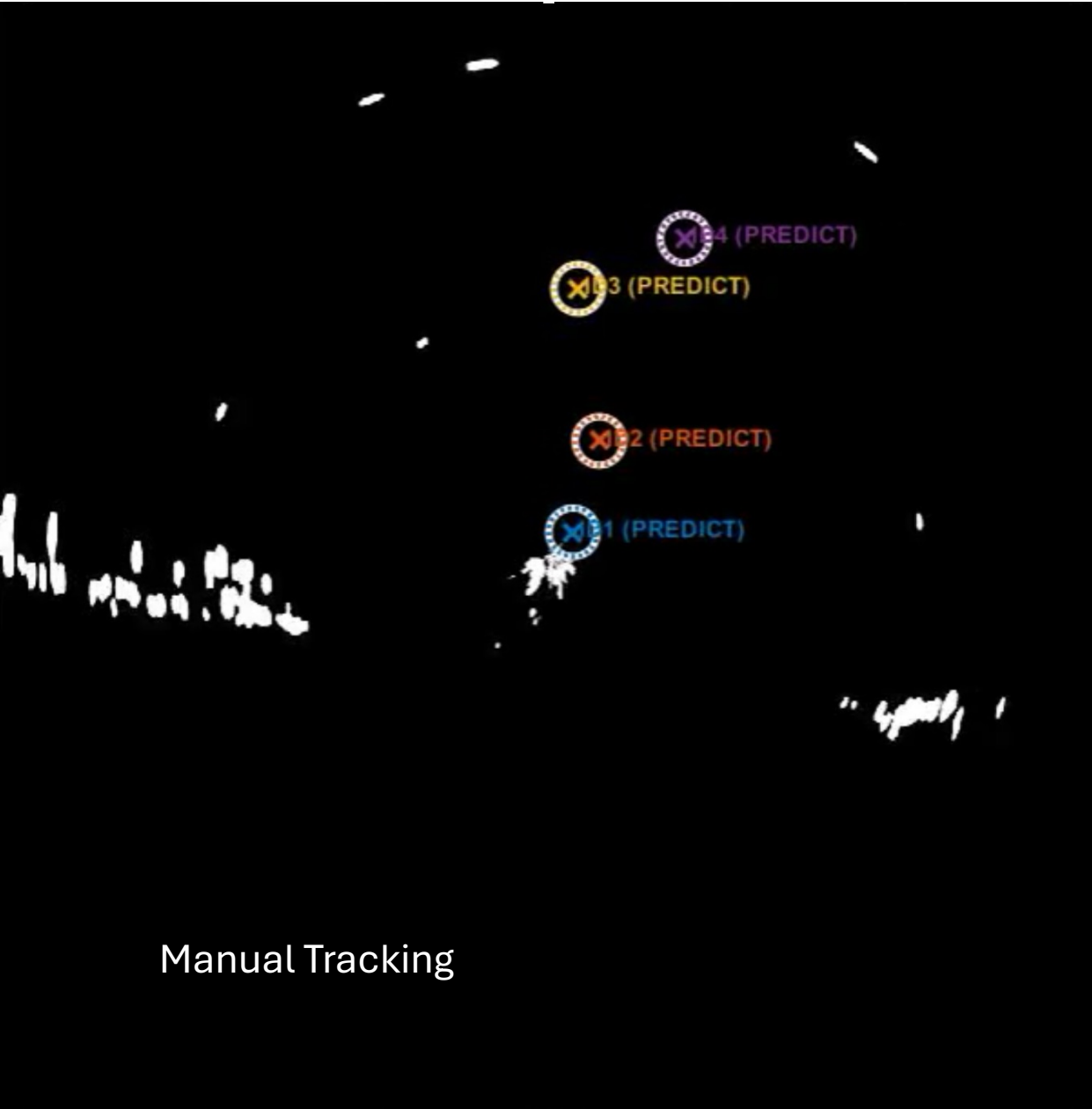
- Non Kalman based object tracking
- Perhaps Large ML models? Apparently work well with unlabeled data and can have emergent abilities

Da End









MAE Tracking(214px)



Manual Tracking(1735px)

